

# Persistence of Community Bank Branches and Redlining: The Case of Minority Depository Institutions

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## Abstract

I study the present-day proximity of urban community banks, Minority Depository Institutions (MDIs) to historically segregated urban neighborhoods. Using a Poisson Generalized Estimating Equation (GEE) framework, I investigate whether MDIs are more likely to exist in “redlined” census tracts based on the Home Owners’ Loan Corporation’s (HOLC) “residential security maps” in 1930s. I find the expected number of Black MDIs increase 10 percent, whereas those of non-MDI banks decrease 1 percent for each 10 percent increase in the tract’s redlined area. I also find Asian MDIs and Hispanic MDIs locations are negatively correlated with the tract’s redlined areas. Rather, their expected numbers increase 34 percent and 13 percent for each 10 percent increase in Asian language and Spanish speakers at the tract level respectively. The results show how historical federal policies continue to influence the modern banking landscape and how Black-owned banks co-exist with larger banks in their local markets.

**JEL Classification Codes:** G20, G21 , L10

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# 1 Introduction

New Deal era agencies, the Home Owner’s Loan Corporation (HOLC) and the Federal Housing Administration (FHA) <sup>1</sup> were created to facilitate the refinancing and insuring of nonfarm residential mortgages after the Great Depression and in the late 1930s, the HOLC created color-coded “residential security maps” evaluating neighborhoods in more than 200 cities. Urban neighborhoods coded green (designated as A) were considered “new, homogeneous, and in demand as residential locations in good times and bad times.”, while areas coded blue (designated as B) were termed “reached their peak”, but were “still desirable”. Finally, areas coded yellow (designated as C) were termed as “definitely declining” and those coded red (designated as D), the lowest rating, were considered as the most “hazardous” (Jackson, 1985) for investment. At the time, a majority of Black neighborhoods were rated hazardous for mortgage lending and received a corresponding Grade D and hence, redlined.

The maps exacerbated mortgage activity in urban neighborhoods that were already impoverished (Jackson, 1985). The construction of these maps were strongly reliant on private realtors and appraisers for neighborhood information (Winling and Michney, 2021) and therefore the maps further heightened the cost of obtaining private credit in these neighborhoods (Hiller and Knowles, 2002). Even prior to the HOLC color coded maps, the FHA evaluated neighborhoods using block level information provided by the New Deal relief programs and the US Census. The agency did not insure mortgages in low income neighborhoods in the cities, where the vast majority of urban Black Americans lived (Fishback et al., 2024).

The effects of the maps persists in the present. Aaronson et al. (2021) find the HOLC maps had consequential and lasting effects on the development of urban neighborhoods in the following decades through reduced credit access, lower home-ownership rates, reduced home values and rent in these neighborhoods while also eroding the wealth of their residents, due to falling property values. Woods (2012) show the maps systematically disadvantaged low-income and minority urban residents and lead to disproportionately substandard housing

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<sup>1</sup>established in 1933 and 1934 respectively

conditions for these communities. Hernandez (2009) studies Sacramento County in California and show a substantially large portion of mortgage application denials, along with subprime mortgage borrowers and property foreclosures during the Great Recession were concentrated in urban regions that were previously designated as “declining” and “hazardous” and graded D by the HOLC maps. Hynsjö and Perdoni (2022) use census tract level data from 1960 until 2010 and find substantially lower property values, home values and home ownership rates in neighborhoods with the lowest grade with the negative effects on property values persisting until the early 1980s. Joshi et al. (2024) use data from 2000 until 2018 find similar results.

This paper studies the locations of depository institutions, known as Minority Depository Institutions (MDIs) and examines whether these institutions exist in greater numbers within neighborhoods that were previously redlined compared to non-MDIs. Economic and social outcomes of local communities are strongly correlated with their ability to access credit. A reliable avenue towards credit accessibility may be as simple as how close a household or small business is to a bank branch (Nguyen, 2019; Herpfer et al., 2022). MDIs are US banks owned and operated primarily by minorities and established to serve minority borrowers that historically had restricted access to credit (Office, 2006). The Federal Deposit Insurance Corporation (FDIC) defines MDIs as banks serving primarily in minority-populated areas and with a majority of the board of directors of minority origin or banks with at least 51% minority ownership. MDIs represent a very small percentage of depository institutions in the US.<sup>2</sup>

In the paper, I also investigate why Black-owned banks (or Black MDIs), in spite of their small numbers, tend to disproportionately locate within previously redlined neighborhoods and the unique roles they play within their immediate neighborhoods. I compare Black-owned banks with other banks within the census tract or neighboring tracts and with non-MDIs in census tracts that have relatively high Black populations but are not served by any

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<sup>2</sup>MDIs consist of 2.8 percent of all depository institutions in the US and the combined assets of \$330 billion of the 147 MDIs (in 2024) represents just 1% of U.S. bank assets (Barth and Xu, 2020; Toussaint-Comeau and Newberger, 2023).

Black-owned bank. These are census tracts with an average of 60% Black population. I show Black-owned banks tend to exist in urban neighborhoods with fewer community banks where the banking landscape is dominated by larger non-community banks. Using Welch's t-tests, I show that Black-owned banks tend to have a greater proportion of their branches in redlined census tracts compared to non-MDIs and consequently, their branches also have a higher deposit share at the census tract-level when compared to non-MDI bank branches in similarly redlined census tracts.

I further argue Black-owned banks do not survive in census tracts where the local Black communities are well integrated within the existing banking system. I hypothesize that the persistence of Black-owned banks in redlined regions are partially results of historical federal policies. These banks formed within neighborhoods that were not well-integrated to the broader banking landscape. Even after the HOLC maps fell out of use by the 1940s, Black-owned banks continue to locate in these neighborhoods. I also show that the Black-owned bank branches that currently locate in these redlined neighborhoods are older (more than 30 years old) small-sized community banks, with very high deposit share at the census tract level. Finally, I also provide a visual and descriptive analysis of Black-owned bank branches and highlight the unique role these banks play in their immediate neighborhoods.

To conduct the research, I construct panel datasets using National Bureau of Economics Research's (NBER) Tract Distance Database to locate neighboring tracts, FDIC's Summary of Deposits, HOLC redlining scores and percentages of different grades ("A, B, C, D") within the census tracts under study and their neighboring tracts and the 5 year American Community Surveys (ACS). I use data on historic HOLC redlining scores for the 2010 census tracts developed by Meier and Mitchell (2023) using percentages of HOLC areas in each tract.<sup>3</sup> Higher HOLC scores reflect lower graded neighborhoods, with a score of four indicating the entire census tract was redlined. The outcome variables are the number of Black-owned banks, Asian-owned banks, Hispanic-owned banks and non-MDIs in the studied tracts. I

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<sup>3</sup>The data is hosted on the Inter-university Consortium for Political and Social Research's (openICPSR) website.



estimate the expected number of bank branches using Poisson Generalized Estimating Equations (GEEs) following Wang et al. (2024).

I find the expected number of Black MDIs increases by 10 percent, whereas the expected number of non-MDI banks decreases by 1 percent for each 10 percent increase in the census tract's area with Grade D. I also find Asian MDIs and Hispanic MDIs locations are either not correlated or negatively correlated with the census tracts' graded D areas. Instead, the expected number of Asian MDIs and Hispanic MDIs increases by 34 percent and 13 percent for each 10 percent increase in the Asian language and Spanish speakers at the census tract level respectively. To understand whether both non-MDIs and Black-owned banks have propensities of locating within very similar tracts, I also control for the median household income at the census tract level. I find the expected number of Black-owned, Asian-owned and Hispanic-owned banks decreases by about 3-7% for each \$10,000 increase in the median household income, whereas the expected number of non-MDIs increase by about 0.5 percent for each \$10,000 increase in the median household income.

As robustness checks, I use the HOLC score as the primary explanatory variable in separate estimations. The results are largely similar. I find the expected number of Black MDIs increases between 13-15 percent, whereas the expected number of non-MDI banks decreases by about 2-3 percent for each point increase (on a four point scale) in the census tract's HOLC scores. In other words, holding all other variables constant, Black-owned banks are more likely to locate in tracts with more redlining. I also find Asian MDIs and Hispanic MDIs locations are again either not correlated or negatively correlated with the HOLC scores.

This is the first paper to study the locations of Black-owned banks and their proximity to historically segregated urban neighborhoods. The findings enrich three areas of existing scholarship. First, the paper contributes to literature studying locations of bank branches. Chang et al. (1997) find bank branches have a tendency to spatially cluster and attribute this behavior to rational herding by banks. Bank branches tend to open in locations with existing branches, even though it leads to less profitability among existing branches. Babajanova

(2022) shows MDIs tend to locate in Low-Moderate Income (LMI) tracts to avoid intense competition but as a result earns less profit when they share a location with a rivalrous MDI or a non-MDI branch.<sup>4</sup>

Second, the paper extends prior literature on the historical segregation of urban areas, racial wealth gap and credit access among minority communities (Aaronson et al., 2021; Hernandez, 2009; Park and Quercia, 2020; Hynsjö and Perdoni, 2022). And finally, the paper contributes to research studying minority-owned banks and their impact within their immediate local communities. Breitenstein et al. (2014) and Eberley et al. (2019) show the markets served by MDI offices include a higher share LMI census tracts, as well as a higher share of minority populations. Similarly, Hurtado and Sakong (2023) show minority borrowers applying for mortgages in banks sharing minority identity are nine percents more likely to be approved than otherwise identical minority borrowers in non-minority banks. Similarly, Razzak (2025) studies closures of both MDI and non-MDI branches within census tracts and demonstrates closing MDI branches generally leads to credit contraction among communities sharing minority identity with the MDI owners whereas closing non-MDI branches more often than not leads to a credit expansion within the local communities.

## **2 Context for Black-owned banks**

### **2.1 Historical context**

Even prior to the existence of the HOLC maps, population sorting based on race was a common phenomenon in major US urban areas. Shertzer and Walsh (2019) uses district-level spatial dataset of major US cities, spanning the years 1900 through 1930, and show a quantitatively large flight of White residents from urban neighborhoods in response to Black arrivals from the South. This phenomenon accelerated between 1900 and 1930. Akbar et al. (2025) use city block-level data of matched addresses, and find that over a single decade, (between 1930 and 1940) rental prices soared by roughly 50% on city blocks that transitioned

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<sup>4</sup>The Department of Housing and Development (HUD) considers U.S. Census Tracts in which 51 % or more of the households earn less than 80 percent of the county’s or metropolitan area’s median income as LMI tracts.

from all White neighborhoods to majority Black neighborhoods, while, home values fell by 10% relative to blocks that maintained all White neighborhoods.

Increasingly segregated, predominantly Black neighborhoods also faced a compounding problem of reduced access to financial institutions, such as banks. This is because, the New Deal included provisions that encouraged the establishment of smaller community banks over larger banking conglomerates or national banks. The decision to advocate for small community banks over larger bank networks was not racially motivated but was implemented to protect small banks from both bank runs and competition of larger conglomerates. Nevertheless, these decisions negatively impacted the prospects for Black-owned banks. Before the implementation of the New Deal, it was generally acknowledged that for Black-owned banks to survive, they needed to be branches of larger national banks and cease to be standalone banks. (Baradaran, 2017). Once the banking legislations of the New Deal had passed, Black-owned banks could no longer be part of larger national conglomerates and were left on their own to serve within their immediate communities.

## **2.2 Current institutional context**

There are very few Black-owned banks that survive in the US. In 2011, total number of such institutions are 30 and by the end of the sample period in 2023, 18 remain. At the beginning of the sample period, there are 154 branches spread over 149 census tracts, while at the end of the sample period there are 79 branches. Over 85% of Black-owned banks have at least one branch located in neighborhoods that were part of the HOLC “residential security maps”. Figure 1 through Figure 3 show MDIs (primarily Black-owned banks) and non-MDIs in some of the largest counties: Baltimore city, New York county (Manhattan) and Bronx county. To understand why Black-owned banks are over-represented in these neighborhoods, I isolate the remaining banks in the year 2011 by age. If only relatively older Black-owned banks are in the redlined neighborhoods, these surviving branches are artifacts of past federal policies. However, if newer banks also establish close to these neighborhoods, it points to a more persistent phenomenon.

Figure 4, Panel A and B show banks that are in the scored neighborhoods are likely to have fewer branches and also have comparatively larger asset sizes.<sup>5</sup> These comparatively larger banks with very little branching footprint also tend to survive longer and many of them are still operating in the year 2023. Figure 4, Panel C and D show branch level characteristics of all surviving Black-owned banks. Interestingly enough, only half of Black-owned bank branches are located within census tracts that were part of the “residential security maps”. As expected, the branches that exists in these neighborhoods have higher deposit share, between 50-60 percent of these bank branches have deposit shares greater than 50% at the census tract level. In other words, these branches are well-integrated within their local communities.

To study whether older Black-owned bank branches are over-represented in these neighborhoods, I divide the ages of the branches into three groups: those aged below 30 years, those aged between 30 and 60 years and those over 60 years. I make these divisions based on the timing of important federal banking legislations that could have impacted branching decisions such as the Fair Housing Act of 1968 and the Riegle–Neal Interstate Banking and Branching Efficiency Act (IBBEA) of 1994. The Fair Housing Act, also known as Title VIII of the Civil Rights Act of 1968, prohibited discrimination in housing based on race, religion, national origin, and sex. While the IBBEA was the primary federal legislation that allowed banks to branch across state lines, establishing a national framework for interstate banking and branching. As expected, Figure 4, Panel D shows a majority of branches within census tracts that were part of the “residential security maps” are older, established prior to the passing of IBBEA.

In order to fully separate the role of Black-owned banks in the recent decade, I separately study two mutually exclusive samples: (a) other banks<sup>6</sup> within census tracts with Black-owned banks along with those in their neighboring tracts (I call it Sample A) consisting of 1107 tracts and (b) non-MDI banks in census tracts with at least 40% Black population but

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<sup>5</sup>About a third of the Black-owned banks in the sample have assets less than 100\$ million.

<sup>6</sup>these banks could be other MDIs and non-MDIs

not served by Black-owned banks (I call it Sample B) consisting of 1023 tracts. In the year 2023, about 14.4% of U.S population identified as Black, so the latter sample would have more than double the average US Black population. Even though there are only 149 census tracts with Black-owned banks in my sample, there are 1023 census tracts with at least 40% Black population but no Black-owned banks.<sup>7</sup> Therefore, a majority of Black populated census tracts in my sample are not served by Black-owned banks.

I take a two pronged approach. First, I compare the institutional structure and branch composition of each set of banks (other banks in Sample A and non-MDIs in Sample B). Second, I compare the loan and deposit activities between Black-owned banks and their neighboring banks (Sample A) and between Black-owned banks and non-neighboring non-MDIs (Sample B). To differentiate the role of Black-owned banks within their immediate communities, I compare the loan and deposit activities of Black-owned banks and banks from Sample A and Sample B. I calculate and compare (a) large-sized loans: residential, commercial real estate (CRE), commercial and industrial (C&I) and small business loans (SBLs); (b) personal loans: credit cards, auto-loans and other personal loans including student loans; and (c) total deposits: individual deposits, public funds and wholesale funds of each of these banks. Loans and deposits are all normalized by asset sizes of the banks.

### 3 Data

I construct panel datasets by combining publicly available datasets. These are the National Bureau of Economics Research’s (NBER) Tract Distance Database to locate neighboring tracts within 1 mile and 5 miles of the census tracts under study, FDIC’s Summary of Deposits (SoD), Federal Financial Institutions Examination Council’s (FFIEC) Call Reports, HOLC’s redlining scores and percentages of different grades within the census tracts and the 5 year American Community Surveys (ACS) resulting in datasets of all census tracts with bank branches within counties where an MDI branch is present in the year 2011. The dataset

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<sup>7</sup>I delete census tracts that are neighboring tracts of Black-owned banks from Sample B to create spatial separation between the two samples.

consists of 12,953 tracts observed over 13 years, 2011 until 2023. Among them about 1090 census tracts contain MDIs. All dollars values are converted to the 2021 dollar value using the Consumer Price Index (CPI) from US Bureau of Labor Statistics (2021) website. I also construct two sets of data, one with features of all neighboring census tracts within a mile of the census tracts under study and one with features of all neighboring census tracts within 5 miles of the census tracts under study.

### 3.1 Redlining Percentages and Scores

I collect redlining proportions for census tracts from two sources. Meier and Mitchell (2023) and the University of Richmond’s *Mapping Inequality* data containing the HOLC spatial data cross-walked into 2010 and 2020 census tracts. From the former data set, I can obtain HOLC scores and from the latter, I recover percentages of redlined or Graded D areas mapped into 2010 census tracts. Meier and Mitchell (2023) classifies 12,834 urban census tracts representing 142 core-based statistical areas (CBSAs) across US based on the HOLC graded maps. They drop census tracts in which less than 20% of the land area was graded by HOLC. They then multiply HOLC graded percentages in each census tracts by number values 1-4, representing each HOLC grade. For example, if 50% of the census tract’s graded area was classified “A” and 50% was “B,” the census tracts receives a score of  $((1 * 0.50) + (0.50 * 2)) = 1.50$ . Lower historical redlining scores indicate higher grades, while higher scores up to 4.0 means almost the entire census tract area was considered hazardous and hence redlined.

### 3.2 Summary of Deposits

I obtain information on bank branches from the SoDs published on the FDIC website. The SoDs contain information on every operating bank branch in the US, including street addresses, zip codes, geolocation data and individual branch deposits, published yearly at the end of June. Initially I collect a unique identification number for each institution assigned by the FDIC (CERT numbers) of MDIs from 2011 until 2023. Using the CERT numbers, I collect bank branch information in all counties where an MDI is present between the years

2011 and 2023. The SoD data does not contain the census tracts where the branches are located. I use the “pygris” package in Python to geolocate latitude and longitude information of each branch to their corresponding census tracts based on the 2010 Census.<sup>8</sup>

### 3.3 Call Reports

I obtain each bank’s deposit, asset and loan information from the Call Reports on the FFIEC website. I collect loan data such as residential loans, CRE loans, C&I loans and total SBLs of every bank in my sample from 2011 until 2023. I also collect personal loan information: credit card, revolving credit, automobile loans and other personal loans such as student loans. Finally, I collect deposit information, both transactional and non-transactional. I collect information on individual deposits, public funds (deposits from federal or state institutions) and wholesale funds (deposits from other commercial banks).

### 3.4 American Community Survey

I collect census tract characteristics from the 5-year ACS website, such as population, percentage of different racial communities, median household income, median home values for all census tracts in counties with MDIs. Appendix A Table A1 has census tract characteristics. The four panels of Table A1 show MDIs tend to locate in census tracts with higher proportions of the communities sharing their minority identity. Moreover, median household income decreases while renters and vacancies increase as one moves from Grade A to Grade D. Even though I consider non-MDI tracts separately for the Summary Statistics table, many non-MDIs do locate within census tracts with MDIs. Figure A1, Panel A through Panel D show median household income, percent of SNAP beneficiary households, percent of vacancies and percent of renters for different HOLC intervals. The figure shows census tracts with Black-owned banks have lower household income and consequently higher percent of SNAP beneficiary households, vacancies and renters. Figures A2 through A4 show MDI and non-MDI branches superimposed on median household income by census tracts in Baltimore city,

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<sup>8</sup>Many branch locations have missing latitude and longitude data. In case of missing latitude and longitude information, I use the API from Google Maps to obtain latitude and longitude information from street addresses of each branch.

New York county (Manhattan) and Bronx county in 2011. The figures show MDI branches generally locate in neighborhoods with lower median household income.

## 4 Empirical analysis

I consider all census tracts with bank branches in counties with MDIs, for my primary analysis.<sup>9</sup> The outcome variables are the number of different MDIs (Black MDIs, Asian MDIs and Hispanic MDIs) and non-MDIs at the census tract. My primary explanatory variables are the percentage of graded D in each tract. I estimate four sets of equations, where the outcome variables are the number of Black MDIs, Asian MDIs, Hispanic MDIs and non-MDIs respectively. In the model I control for population within the census tract, population in all census tracts within a mile (or 5 miles) radius, percent of the community sharing minority identity with MDI owners, median household income, languages spoken at the tract level and other bank branches. For robustness, I estimate another Poisson GEE using the HOLC scores for each tract.

I estimate the expected number of bank branches using Wang et al. (2024). They introduce the Poisson GEE for estimating longitudinal data. Standard log-linearized models estimated using Ordinary Least Squares (OLS) and Generalized Linear Models (GLMs) can be quite misleading in the presence of heteroscedasticity. Therefore, they are not suitable for data with repeated observations because the former methods assume independence of observations. This independence of observations are violated in longitudinal settings where one observes within-unit correlations. GEEs are generally used to model population-averaged effects considering within-unit correlation. They are typically adopted to estimate the parameters of a generalized linear model when the correlation between repeated outcome observations are unknown (Liang and Zeger, 1986).

In essence, GEEs extend GLMs by introducing a working correlation matrix (e.g., exchangeable, autoregressive, unstructured) to model dependencies between repeated measure-

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<sup>9</sup>In Appendix B, I also extend my sample to include all census tracts within counties with MDI branches. As a result, some of these census tracts do not have a bank branch.



ments and provide robust “sandwich” standard errors.<sup>10</sup> This paper assumes an exchangeable correlation structure, i.e., the correlation between any two measurements within the same unit of observations (census tracts) is time invariant, a reasonable assumption since I consider only thirteen years of observations. An advantage of the GEE specification is as long as the mean function is correctly specified, parameter estimates from the GEE are consistent even if the working variance-covariance structure is misspecified (Wang et al., 2024), which is more generally difficult to pin down correctly.

The conditional mean is given by

$$E(y_{i,t}|\mathbf{X}_{i,t}, w_i) = \exp(\alpha_0 + \beta w_i + \beta^T \mathbf{X}_{i,t}) + \nu_{i,t}. \quad (1)$$

where  $w_i$  is the primary explanatory variable, either the percent of graded D areas at the tract level or the HOLC score in census tract  $i$  and  $\mathbf{X}_{i,t}$  is a vector of other controls at the census tract level.

$$\begin{aligned} \mathbf{X}_{i,t} = [ & \textit{Population}, \textit{Population } x \textit{ mile}, \textit{Percent sharing minority identity}, \\ & \textit{Percent sharing minority identity } x \textit{ mile}. \\ & \textit{Median household income}, \textit{Other bank branches}] \end{aligned}$$

In equation (1),  $y_{i,t}$  is the conditional mean of the outcome variable is the number of different MDIs (Black MDIs, Asian MDIs and Hispanic MDIs) and non-MDIs at the census tract. The percentage increase in the expected number of MDIs or non-MDIs is calculated using the following equation.

$$\frac{E(y_{i,t}|\mathbf{X}_{i,t}, w_i = (w_i + V)) - E(y_{i,t}|\mathbf{X}_{i,t}, w_i = (w_i))}{E(y_{i,t}|\mathbf{X}_{i,t}, w_i = (w_i))} \times 100 \quad (2)$$

Using equation (2), I recover the expected change in the number of different kinds of MDIs and non-MDIs based on a unit change in my explanatory variables. I use increments of 10 percent increase as the unit change in the case of  $w_i$  (represented by  $V$ ) and \$10,000 for

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<sup>10</sup>“Sandwich” standard errors account for misspecification of the correlation structure.

a unit change in median household income. The results or their significance do not change based on the unit change used.

## 5 Results

This section provides estimates from the Poisson GEE estimation models. As discussed in the previous section, I use the percent of the census tracts with Grade D and the HOLC scores as the primary explanatory variables. Table 1 through Table 4 show the results. In Table 1 and Table 3, neighboring tracts are those within 1 mile of the studied tracts whereas in Table 2 and Table 4, neighboring tracts are those within 5 miles of the studied tracts. The results are consistent across the different specifications.

### 5.1 Results using census tracts' redlined or graded D areas:

Table 1 and Table 2 show estimates from the Poisson GEE model using percent of the census tracts with Grade D as the primary explanatory variable. I find the expected number of Black-owned banks to increase by about 10 percent and the expected number of non-MDI bank branches to decrease by 1 percent for each 10 percent increase in census tract area with Grade D. I also find Asian-owned and Hispanic-owned banks locations are not positively influenced by the census tracts' Grade D areas. The results hold when I consider neighboring tracts within both 1 mile and 5 miles of the studied tracts. Conversely, I find Asian-owned and Hispanic-owned banks' locations are strongly influenced by the languages spoken at the census tract level. Notably, the expected number of Asian MDIs and Hispanic MDIs increases by 34 percent and 13 percent for each 10 percent increase in the number of Asian language or Spanish speakers within the census tracts respectively.

### 5.2 Results using census tracts' HOLC or redlining scores

Table 3 and Table 4 show estimates from the Poisson GEE model using the HOLC or redlining score variable. The results are largely similar to Table 1 and Table 2. The expected number of Black-owned banks increases between 13-15 percent, whereas the expected number of non-MDI banks decreases 2-3 percent for each point increase in the census tracts' HOLC

scores. In other words, holding all other control variable constant, including the number of non-MDIs branches, Black-owned banks are more likely to locate in census tracts with higher scores. I also find Asian-owned and Hispanic-owned banks locations are not positively influenced by the census tracts' HOLC scores. The results are consistent when I consider neighboring tracts within both 1 mile and 5 miles of the studied tracts. Again, I find the expected number of Asian MDIs and Hispanic MDIs increases by 37 percent and 10 percent for each 10 percent increase in the number of Asian language or Spanish speakers within the census tracts respectively. The results are consistent across all specifications.

### 5.3 Results with Welch's t-tests

I compare the proportion of bank branches (for each Black-owned bank) in primarily redlined census tracts vs. proportion of bank branches (for each non-MDI) in primarily redlined census tracts. The results are provided in Table 5. The mean Black-owned bank has about 25 percent of its branches in redlined tracts whereas the mean non-MDI has about 5 percent of its branches in the redlined tracts. The results gives validation to the GEE estimation results shown in Tables 1 through 4. I also test deposit share among Black-owned bank branches vs. non-MDI branches in primarily redlined census tracts. Black-owned banks contain a higher deposit share when compared to non-MDIs in these census tracts. Finally, I compare the ages of bank branches and compare Black-owned banks and non-MDI banks in redlined tracts. Even though on average, Black-owned bank branches tend to be older, the results are not statistically significant.

In Table 6, I compare age composition and deposit share of Black-owned bank branches in redlined census tracts vs. census tracts that either were not graded or received grades of A, B and C. I find no significant differences between the means of these two sub-samples. Although, bank branches in redlined tracts are on average slightly older, the differences in their means are not statistically significant. I also compare the age composition and deposit share of non-MDI bank branches in redlined census tracts vs. census tracts that either were not graded or received the grades of A, B and C. Non-MDI bank branches that are

currently present in redlined census tracts are significantly younger. The results imply one of two things, either these non-MDI branches were more recently established or older non-MDI branches do not survive in these tracts for long periods of time.

#### **5.4 Results using all census tracts with or without bank branches**

In order to rule out that my results are driven by sample selection, I do a second set of analysis. The results are provided in Appendix B. Here, the sample includes all census tracts in counties with MDIs, whether the census tracts have bank branches or not. In Appendix B, I use percentages of areas with Grade D within the census tracts as the primary explanatory variable. I find the expected number of Black-owned banks increase by 6-7 percent for each ten percent increase in Grade D areas within the census tracts. Meanwhile, the expected number of non-MDIs decrease by 2-3 percent for each ten percents increase in areas with Grade D within their neighboring tracts. The results are similar when I use the HOLC scores of the census tracts.

The overall results in Appendix B are largely similar to the results in my primary analysis. One exception is when I use the latter sample, for every \$10,000 increase in household median income, the expected number of non-MDIs in a census tract decreases by 1 percent. This result can be explained by the nature of the sample. Since I consider census tracts in primarily large urban counties, bank branches do tend to locate more often than not within census tracts near downtown areas. Many of these downtown locations also have more low-income households. Compare this set of results with those from Table 1 through Table 4, where the sample is all census tracts with at least one bank branch. As expected, in this set of results, the expected number of non-MDIs increase by one percent for each \$10,000 increase in household median income.

#### **5.5 Results using average graded D or HOLC scores of neighboring tracts**

In Appendix C, I check whether the presence of Black-owned banks are also driven by the percent graded D tracts in neighboring census tracts. Here, I use the average HOLC scores within a mile radius (or 5 mile radius) and average graded D tracts with a mile (or 5 mile

radius) of MDI containing tracts as the primary explanatory variables. Again, the results are largely similar to my primary estimation results.

I find the expected number of Black-owned banks increase by 13-16 percent for each ten percent increase in graded D areas within their neighboring tracts. Meanwhile, the expected number of non-MDIs also increase by 3-7 percent for each 10 percent increase in the census tracts' area with Grade D. As robustness checks, I also use the HOLC scores of the census tracts as the primary explanatory variable. Again, I find the expected number of Black-owned banks increase 14-18 percent for each point increase in the average HOLC scores of their neighboring tracts and the expected number of non-MDIs increase 2-6 percent for each point increase in the average HOLC scores of the census tracts.

These results are not surprising as many non-MDIs in the sample are within census tracts that were part of the “residential security maps” but were graded A or B. The results, therefore, show a degree of spatial clustering among Black MDIs, Asian MDIs, Hispanic MDIs and non-MDI bank branches. Figure 1 through Figure 3 confirms this observation. The HOLC maps contained contiguous regions of the highest grade (Green for Grade A) and the lowest grade (Red for Grade D). Hence the results that non-MDI bank branches tend to locate within a mile or 5 miles of the lowest graded neighborhoods are unsurprising.

## **6 Are Black-owned Banks Different?**

To understand why Black-owned banks are over-represented in redlined neighborhoods, I study bank and branch level characteristics of non-MDI banks present in majority Black communities, farther away from Black-owned banks. These banks are not officially designated as Black-owned banks or MDIs. At the beginning of the sample period in 2011, 80% of these banks had asset sizes less than \$10 billion, hence a majority of them were small-sized community banks. At the end of the sample period in 2023, these small-sized community banks consist of 65% of the sample. Community banks in this time-frame went through several phases of consolidation (Minton et al., 2024) and it is possible that some existing banks merged and became larger. About two-thirds of these non-MDI bank branches are

more than 30 years old and about a third are sixty or older. More than 70% of the banks operate a maximum of two branches in the studied tracts, and 86% of the banks operate a maximum of six branches. The percentage remain constant between 2011 and 2023, even though many banks close or consolidate.<sup>11</sup> About 50% of branches holds a deposit share greater than 50% within the census tracts. In other words, the branches are well-integrated and specialized in serving their immediate communities.

Figure 5, Panels (a) through (d) compare neighboring banks of Black-owned banks in previously redlined tracts (Sample A) with the non-neighboring non-MDI banks serving in areas with at least 40% Black population (Sample B). What is immediately striking are the number of banks and branches that serve the latter population, even though the sample number of census tracts are similar, 1107 vs. 1023 census tracts. The number of banks and corresponding branches are much higher for Sample B in the year 2011, although the two sample converge in numbers towards the end of the sample period in 2023. At the beginning of the sample period, these non-MDI banks tend to be small-sized and a majority of these banks can be considered as community banks based on FDIC's definition of community banks (FDIC, 2012). Towards the end of the sample period, older and relatively larger banks with greater deposit share at the census tract level survive. Notably, Figure 5 shows banks in Sample A are larger banks with a majority of branches holding less than 20% of the deposits at the census tract level.

To further investigate the role of Black-owned banks in their immediate neighborhoods, I compare bank level loan activity and deposit data. My first comparison is between Black-owned banks and other banks in their immediate neighborhoods in Sample A.<sup>12</sup> My second comparison is between Black-owned banks and non-neighboring non-MDI banks in Sample B. The non-MDIs in Sample B serve in census tracts with significant Black populations. A visual comparison between Black-owned banks and Sample A and Sample B is provided in

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<sup>11</sup>The caveat here is I only see what is happening at the local level, they may possibly have other branches in other counties or states

<sup>12</sup>The comparison sample is all non-Black-owned banks in census tracts with Black MDIs and their immediate neighboring census tracts.

Figures 6 and 7.

When compared to other banks operating within the same tracts or neighboring tracts, shown in Figure 6 panels (a) through (c), Black-owned banks tend to provide higher proportions of residential and CRE loans. However, they provide much lesser quantity of C&I loans and SBLs. In other words, Black-owned banks specialize in providing loans for residential purposes (both mortgages and to landlords) in their immediate communities. Black-owned banks also provide fewer credit cards and other revolving credit and automobile loans. In recent years, they have offered more personal loans, which also includes student loans. Similarly, when I compare Black-owned banks with non-MDI banks that serve communities with significant Black populations, but are not designated as Black-owned banks or MDIs, shown in Figure 7 panels (a) through (c), the general overview remain the same. Interestingly enough Black-owned banks also hold more public funds as deposits compared to other kinds of banks.

## 7 Conclusion

In this paper, I demonstrate the prevalence of Black-owned banks in previously redlined urban neighborhoods. I use Poisson GEEs to estimate the increase in the expected numbers of Black-owned, Hispanic-owned, Asian-owned and other non-MDI banks in their respective census tracts with an increase in the tracts' graded D areas. I find the expected number of Black MDIs increases 10 percent, whereas the expected number of non-MDI banks decreases 1 percent for each 10 percent increase in the census tracts' Graded D area. I also find Asian-owned and Hispanic-owned banks' location are not positively related with the census tracts' Grade D areas or higher HOLC scores. Instead, the expected number of Asian MDIs and Hispanic MDIs increases by 34 percent and 13 percent for each 10 percent increase in the number of Asian or Spanish speakers within the census tracts respectively. For robustness, I use the HOLC or redlining scores as the primary explanatory variable. The results are consistent across different kinds of specifications.

To provide an explanation for the observed phenomenon, I draw on historical literature

focusing on the interaction between the HOLC maps, the residents of the redlined neighborhoods, their difficulty accessing credit and how the need for this access lead to the formation of Black-owned banks within these neighborhoods (Hiller and Knowles, 2002; Baradaran, 2017; Winling and Michney, 2021). To understand why Black-owned banks are still over-represented in these neighborhoods, I study individual characteristics of Black-owned banks, including their branch characteristics, and bank-level loan activities in the present. I show Black-owned banks that continue to survive in these neighborhoods tend to be older, with very high deposit share and very little branching footprint. Many of them survive to the end of the sample in 2023, demonstrating some kind of resilience. Using Welch’s t-tests I also show Black-owned banks have a greater proportion of their branches located in redlined census tracts when compared to non-MDIs. At the same time, their branches also have a higher deposit share at the census tract level when compared to non-MDI bank branches in similarly redlined tracts.

In my sample, I also find a large share of banks that are not Black-owned or designated MDIs but serve in primarily Black communities. These non-MDIs integrate well within their local communities and provide a significant share of residential, CRE and C&I loans and SBLs and larger proportions of credit card and automobile loans compared to Black-owned banks. Individually, these banks tend to be small-sized community banks with limited branching footprint. Moreover, more than 50% of these bank branches hold a deposit share greater than 50% at the census tract level, meaning they form an integral part of the banking services accessible to these primarily Black minority communities.

Finally, I show Black-owned banks are quite dependent on individual deposits and hold greater quantities of public funds as deposits compared to non-MDIs. Holding greater quantities of public funds may protect surviving banks against financial shocks (Eberley et al., 2019). Although Black-owned banks provide a slightly greater share of residential, CRE and personal loans to their immediate local communities, in general, they do not play an out-sized role in providing other kinds of credit. In other words, current Black-owned banks in



redlined neighborhoods specialize in mortgage originations and rental property loans. This is unsurprising, since many of them were historically formed to address higher costs of obtaining credit within the redlined neighborhoods.

I conclude the over-representation of Black-owned banks within historically segregated urban neighborhoods is most likely due to a mix of past federal policies, White out-migration from core urban areas in early 20th century (Shertzer and Walsh, 2019; Akbar et al., 2025), leading to increased segregation and a lack of financial integration of the remaining communities within these neighborhoods to the broader banking landscape. Finally I show, surviving Black-owned bank branches specialize and continue to serve local communities in urban neighborhoods with relatively fewer well-integrated community banks.

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**Table 1:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −6.459***<br>(0.114) | −4.663***<br>(0.040) | −5.409***<br>(0.043) | −0.039<br>(0.027)    |
| Population (000s)                                     | −0.056***<br>(0.015) | 0.019***<br>(0.005)  | 0.005<br>(0.005)     | 0.031***<br>(0.001)  |
| Population (000s) 1 mile                              | 0.007***<br>(0.001)  | −0.001*<br>(0.001)   | 0.006***<br>(0.000)  | 0.001***<br>(0.000)  |
| Percent Graded D                                      | 0.097***<br>(0.008)  | −0.001<br>(0.010)    | −0.059***<br>(0.009) | −0.008***<br>(0.002) |
| Median Household Income                               | −0.046***<br>(0.012) | −0.043***<br>(0.004) | 0.006<br>(0.006)     | 0.005***<br>(0.001)  |
| Other banks' branches                                 | 0.111***<br>(0.004)  | 0.082***<br>(0.003)  | 0.106***<br>(0.003)  |                      |
| Percent Black                                         | 0.374***<br>(0.019)  |                      |                      | 0.006*<br>(0.003)    |
| Percent Black 1 mile                                  | 0.008***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                  | −0.151***<br>(0.024) |                      |                      | 0.091***<br>(0.003)  |
| Percent Asian                                         |                      | 0.327***<br>(0.021)  |                      | 0.036***<br>(0.004)  |
| Percent Asian 1 mile                                  |                      | 0.008***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                |                      | 0.293***<br>(0.023)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.456***<br>(0.017)  | −0.116***<br>(0.003) |
| Percent Hispanic 1 mile                               |                      |                      | 0.009***<br>(0.001)  |                      |
| Percent Spanish speakers                              |                      |                      | 0.120***<br>(0.021)  |                      |
| Percent White                                         |                      |                      |                      | 0.069***<br>(0.003)  |
| Scale                                                 | 0.651                | 1.102                | 1.083                | 2.712                |
| Observations                                          | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within one mile.

**Table 2:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −6.929***<br>(0.111) | −4.937***<br>(0.042) | −5.833***<br>(0.051) | −0.061*<br>(0.027)   |
| Population (000s)                                     | −0.075***<br>(0.015) | 0.021***<br>(0.005)  | 0.015***<br>(0.004)  | 0.031***<br>(0.001)  |
| Population (000s) 5 miles                             | 0.001***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  |
| Percent Graded D                                      | 0.102***<br>(0.008)  | −0.026**<br>(0.010)  | −0.032***<br>(0.008) | −0.012***<br>(0.002) |
| Median Income Household                               | −0.059***<br>(0.013) | −0.054***<br>(0.004) | −0.024***<br>(0.006) | 0.004***<br>(0.001)  |
| Other banks' branches                                 | 0.109***<br>(0.004)  | 0.081***<br>(0.003)  | 0.103***<br>(0.003)  |                      |
| Percent Black                                         | 0.309***<br>(0.018)  |                      |                      | 0.006<br>(0.003)     |
| Percent Black 5 miles                                 | 0.024***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                  | −0.118***<br>(0.023) |                      |                      | 0.085***<br>(0.003)  |
| Percent Asian                                         |                      | 0.196***<br>(0.024)  |                      | 0.039***<br>(0.004)  |
| Percent Asian 5 miles                                 |                      | 0.024***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                |                      | 0.364***<br>(0.024)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.200***<br>(0.023)  | −0.112***<br>(0.003) |
| Percent Hispanics 5 miles                             |                      |                      | 0.030***<br>(0.002)  |                      |
| Percent Spanish speakers                              |                      |                      | 0.089***<br>(0.019)  |                      |
| Percent White                                         |                      |                      |                      | 0.069***<br>(0.003)  |
| Scale                                                 | 0.573                | 1.161                | 1.155                | 2.713                |
| Observations                                          | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$   
 Neighboring tracts are within five miles.

**Table 3:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on HOLC Score using Meier and Mitchell (2023)</i> |                      |                      |                      |                      |
|------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                            | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                                | −6.423***<br>(0.114) | −4.659***<br>(0.042) | −5.348***<br>(0.044) | −0.033<br>(0.027)    |
| Population (000s)                                          | −0.067***<br>(0.015) | 0.019***<br>(0.005)  | 0.002<br>(0.005)     | 0.030***<br>(0.001)  |
| Population (000s) 1 mile                                   | 0.007***<br>(0.001)  | −0.001<br>(0.001)    | 0.008***<br>(0.000)  | 0.001***<br>(0.000)  |
| HOLC Score                                                 | 0.129***<br>(0.022)  | −0.002<br>(0.015)    | −0.235***<br>(0.013) | −0.017***<br>(0.003) |
| Median Household Income                                    | −0.057***<br>(0.012) | −0.043***<br>(0.004) | 0.003<br>(0.006)     | 0.005***<br>(0.001)  |
| Other banks' branches                                      | 0.112***<br>(0.004)  | 0.081***<br>(0.003)  | 0.103***<br>(0.003)  |                      |
| Percent Black                                              | 0.374***<br>(0.019)  |                      |                      | 0.006*<br>(0.003)    |
| Percent Black 1 mile                                       | 0.006***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                       | −0.128***<br>(0.022) |                      |                      | 0.094***<br>(0.003)  |
| Percent Asian                                              |                      | 0.324***<br>(0.022)  |                      | 0.034***<br>(0.004)  |
| Percent Asian 1 mile                                       |                      | 0.008***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                     |                      | 0.293***<br>(0.023)  |                      |                      |
| Percent Hispanic                                           |                      |                      | 0.444***<br>(0.017)  | −0.118***<br>(0.003) |
| Percent Hispanic 1 mile                                    |                      |                      | 0.008***<br>(0.001)  |                      |
| Percent Spanish speakers                                   |                      |                      | 0.131***<br>(0.020)  |                      |
| Percent White                                              |                      |                      |                      | 0.069***<br>(0.003)  |
| Scale                                                      | 0.621                | 1.199                | 1.067                | 2.706                |
| Observations                                               | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within one mile.

**Table 4:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on HOLC Score using Meier and Mitchell (2023)</i> |                      |                      |                      |                      |
|------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                            | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                                | −6.946***<br>(0.114) | −4.913***<br>(0.043) | −5.801***<br>(0.051) | −0.054*<br>(0.027)   |
| Population (000s)                                          | −0.078***<br>(0.015) | 0.020***<br>(0.005)  | 0.013**<br>(0.005)   | 0.031***<br>(0.001)  |
| Population (000s) 5 miles                                  | 0.001***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  |
| HOLC Score                                                 | 0.150***<br>(0.021)  | −0.067***<br>(0.018) | −0.192***<br>(0.013) | −0.026***<br>(0.003) |
| Median Household Income                                    | −0.067***<br>(0.013) | −0.056***<br>(0.004) | −0.030***<br>(0.006) | 0.004***<br>(0.001)  |
| Other banks' branches                                      | 0.111***<br>(0.004)  | 0.079***<br>(0.003)  | 0.101***<br>(0.003)  |                      |
| Percent Black                                              | 0.303***<br>(0.018)  |                      |                      | 0.006*<br>(0.003)    |
| Percent Black 5 miles                                      | 0.023***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                       | −0.118***<br>(0.023) |                      |                      | 0.088***<br>(0.003)  |
| Percent Asian                                              |                      | 0.185***<br>(0.024)  |                      | 0.036***<br>(0.004)  |
| Percent Asian 5 miles                                      |                      | 0.024***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                     |                      | 0.376***<br>(0.025)  |                      |                      |
| Percent Hispanic                                           |                      |                      | 0.206***<br>(0.023)  | −0.116***<br>(0.003) |
| Percent Hispanics 5 miles                                  |                      |                      | 0.029***<br>(0.002)  |                      |
| Percent Spanish speakers                                   |                      |                      | 0.090***<br>(0.019)  |                      |
| Percent White                                              |                      |                      |                      | 0.069***<br>(0.003)  |
| Scale                                                      | 0.552                | 1.136                | 1.125                | 2.704                |
| Observations                                               | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$   
 Neighboring tracts are within five miles.

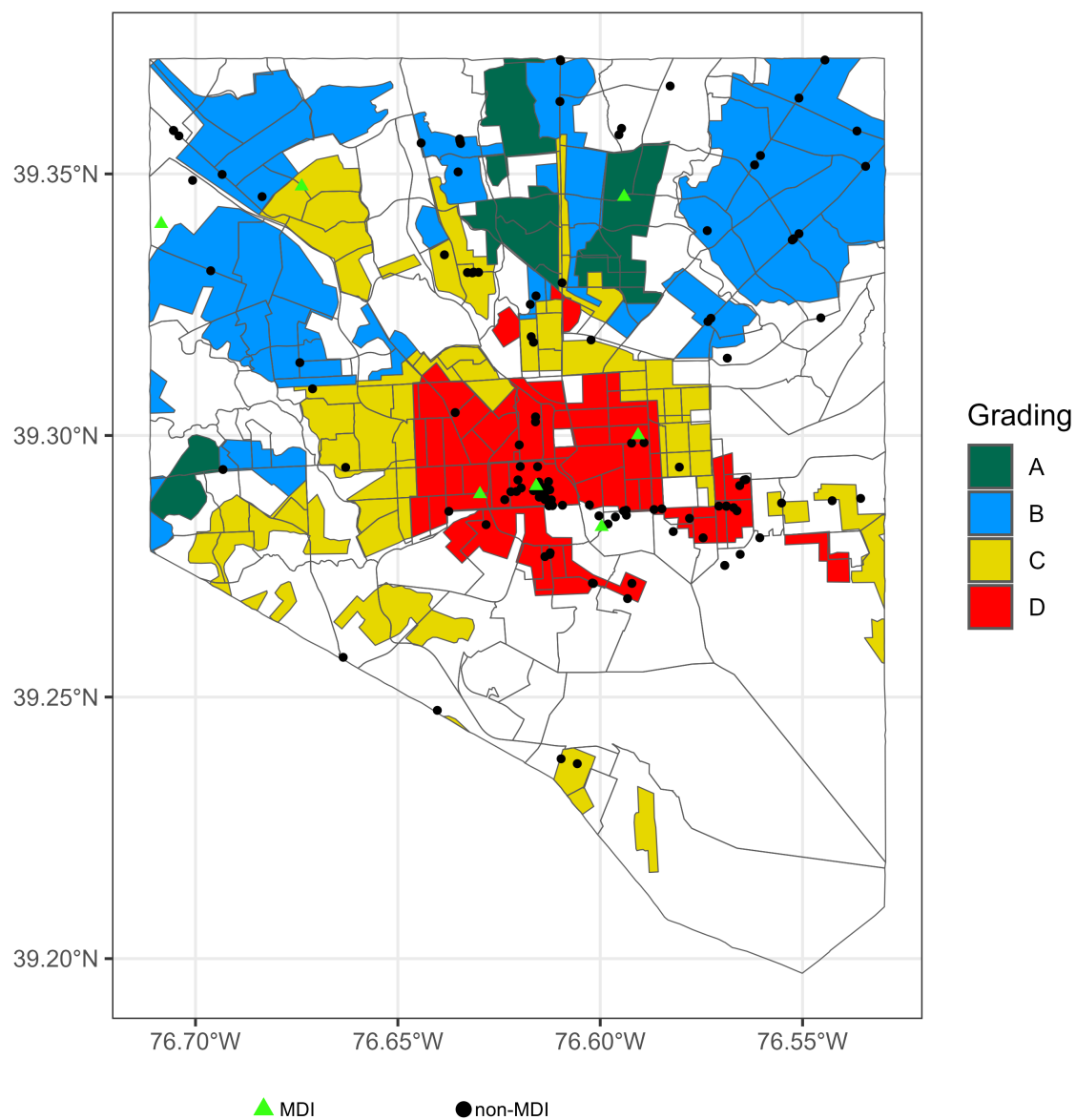


**Table 5:** Two-sample t-test results comparing Black-owned banks and non-MDIs in redlined census tracts

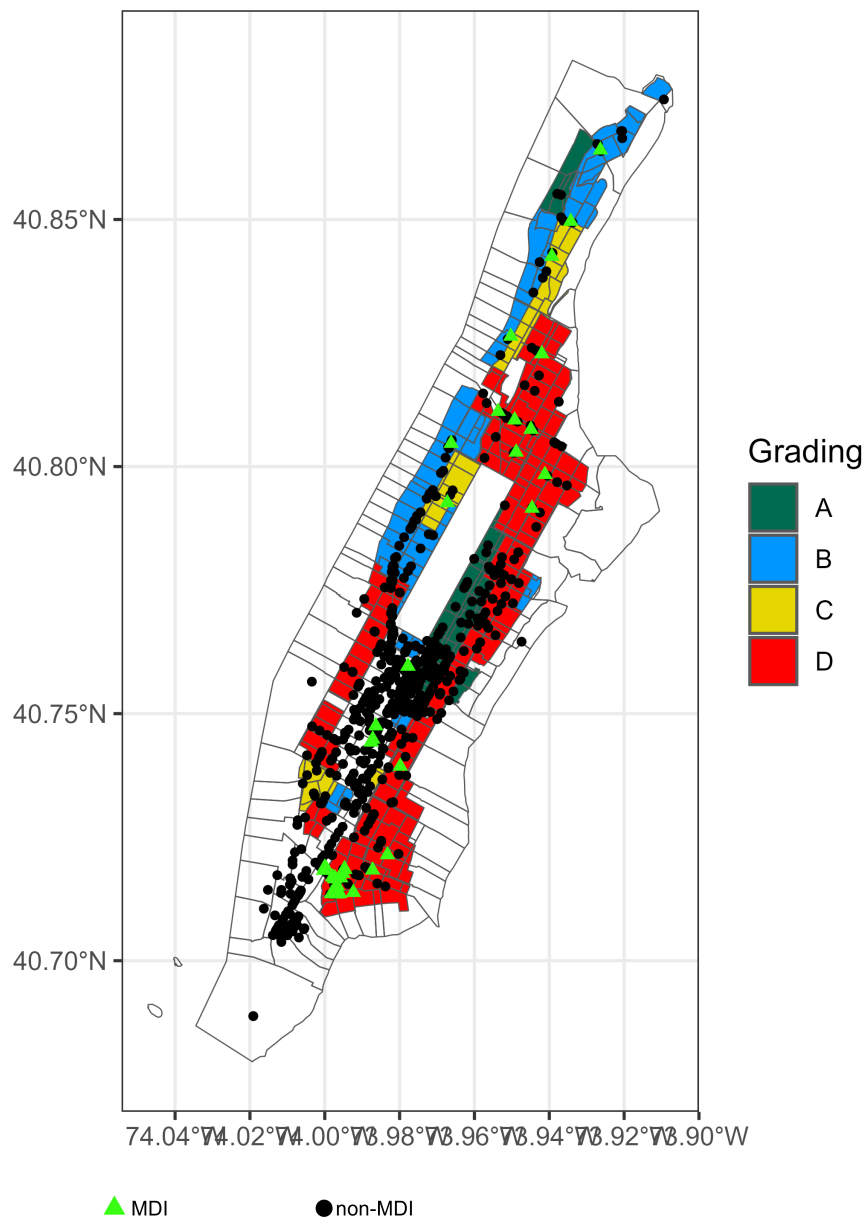
| Panel A: Share of branches in redlined census tracts         |                   |          |                |                    |
|--------------------------------------------------------------|-------------------|----------|----------------|--------------------|
|                                                              | Black-owned banks | non-MDIs | Welch's t-test | Degrees of Freedom |
| Mean                                                         | 0.27              | 0.05     | 4.05           | 29.40              |
| Standard deviation                                           | 0.28              | 0.19     |                |                    |
| Panel B: Deposit-share of branches in redlined census tracts |                   |          |                |                    |
|                                                              | Black-owned banks | non-MDIs | Welch's t-test | Degrees of Freedom |
| Mean                                                         | 54.93             | 40.08    | 1.78           | 19.71              |
| Standard deviation                                           | 35.63             | 33.09    |                |                    |
| Panel C: Age of branches in redlined census tracts           |                   |          |                |                    |
|                                                              | Black-owned banks | non-MDIs | Welch's t-test | Degrees of Freedom |
| Mean                                                         | 66.39             | 56.99    | 1.48           | 23.44              |
| Standard deviation                                           | 29.95             | 44.91    |                |                    |

**Table 6:** Two-sample t-test results comparing Black-owned banks in redlined vs. non-redlined census tracts and non-MDIs in redlined vs. non-redlined census tracts.

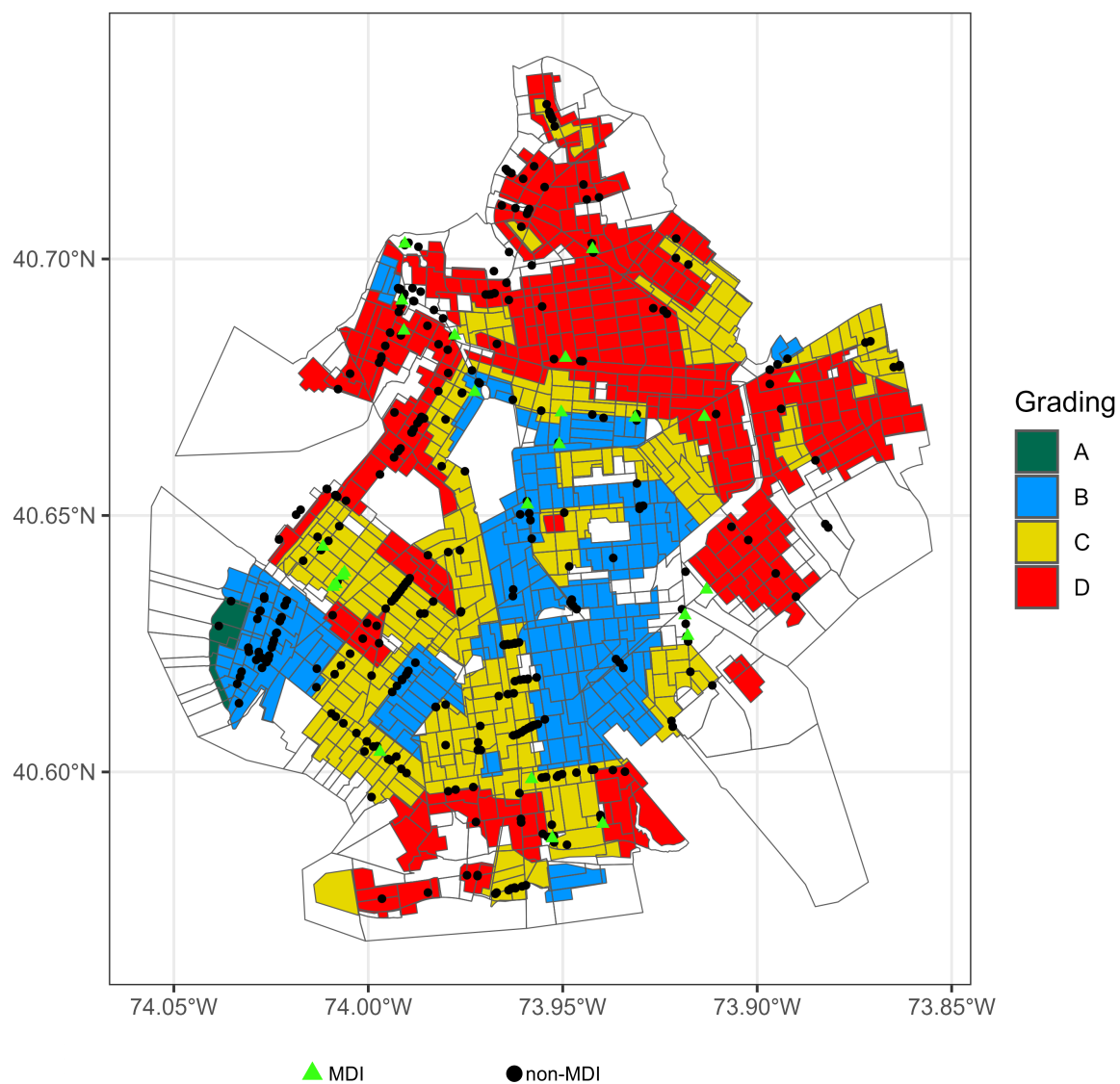
| Panel A: Deposit-share of Black-owned bank branches |          |              |                |                    |
|-----------------------------------------------------|----------|--------------|----------------|--------------------|
|                                                     | Redlined | Non-Redlined | Welch's t-test | Degrees of Freedom |
| Mean                                                | 54.93    | 57.70        | -0.28          | 31.19              |
| Standard deviation                                  | 35.63    | 26.57        |                |                    |
| Panel B: Ages of Black-owned bank branches          |          |              |                |                    |
|                                                     | Redlined | Non-Redlined | Welch's t-test | Degrees of Freedom |
| Mean                                                | 66.39    | 55.25        | 1.18           | 33.61              |
| Standard deviation                                  | 29.95    | 28.25        |                |                    |
| Panel C: Deposit-share of non-MDI branches          |          |              |                |                    |
|                                                     | Redlined | Non-Redlined | Welch's t-test | Degrees of Freedom |
| Mean                                                | 40.08    | 38.63        | 0.77           | 444.97             |
| Standard deviation                                  | 33.09    | 27.14        |                |                    |
| Panel D: Ages of non-MDI branches                   |          |              |                |                    |
|                                                     | Redlined | Non-Redlined | Welch's t-test | Degrees of Freedom |
| Mean                                                | 56.99    | 65.71        | -2.84          | 424.43             |
| Standard deviation                                  | 44.91    | 51.16        |                |                    |



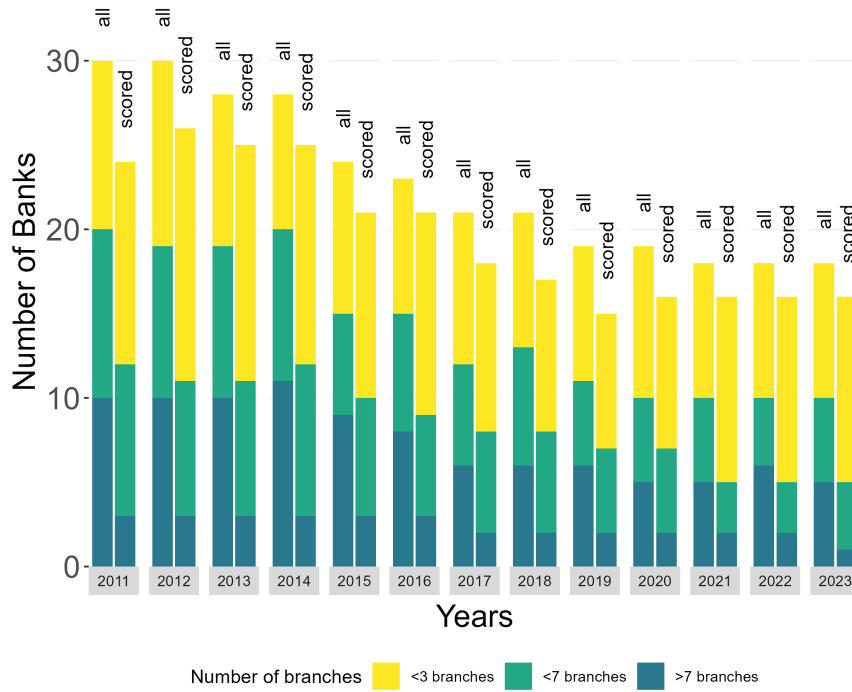
**Figure 1:** The figure shows MDI and non-MDI branches superimposed on the HOLC maps in Baltimore city in 2011.



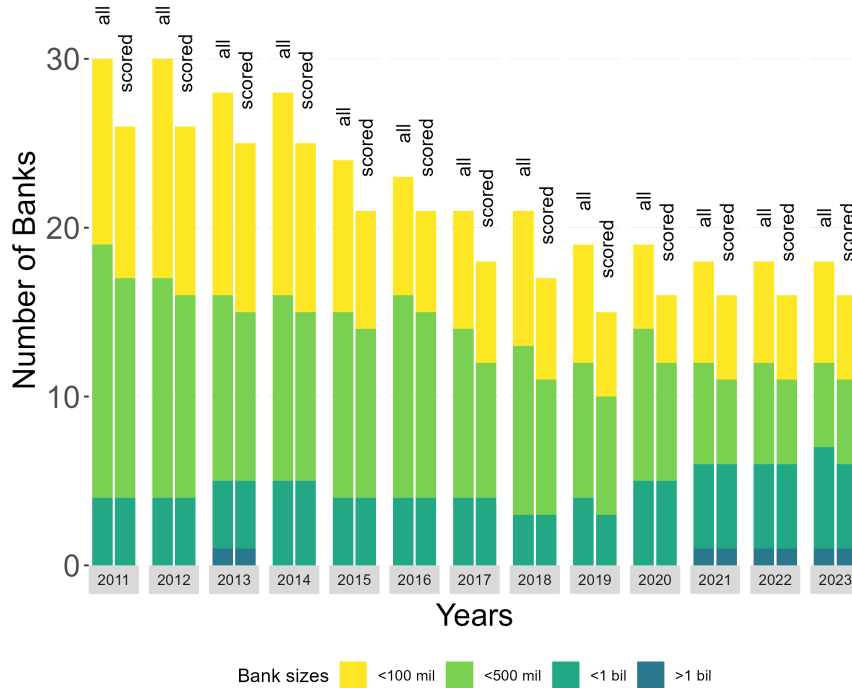
**Figure 2:** The figure shows MDI and non-MDI branches superimposed on the HOLC maps in New York county (Manhattan) in 2011.



**Figure 3:** The figure shows MDI and non-MDI branches superimposed on the HOLC maps in Bronx county in 2011.

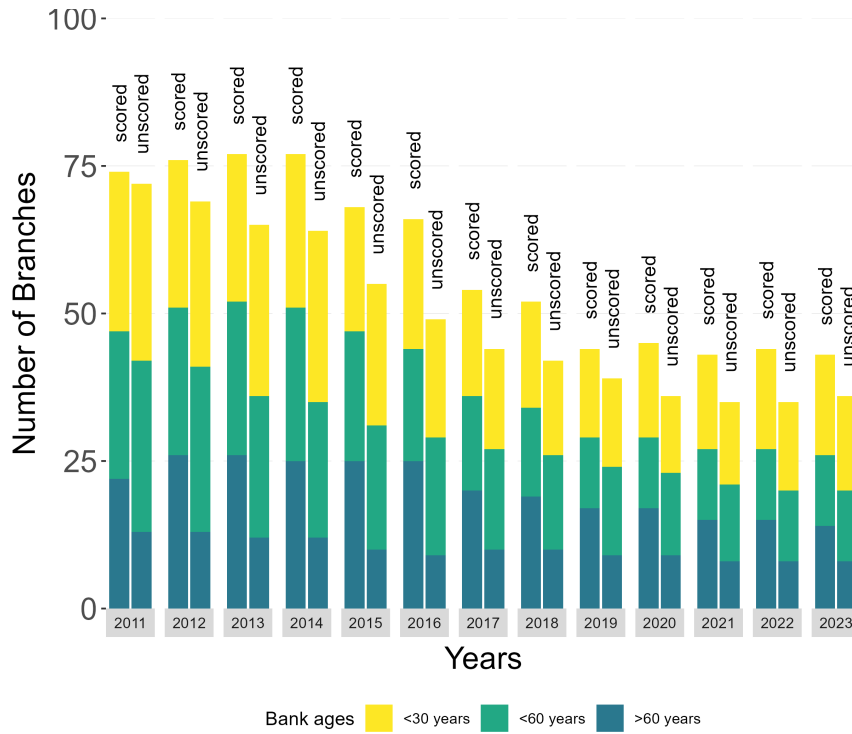


(a) Branch composition among banks in all census tracts vs. those in scored census tracts.

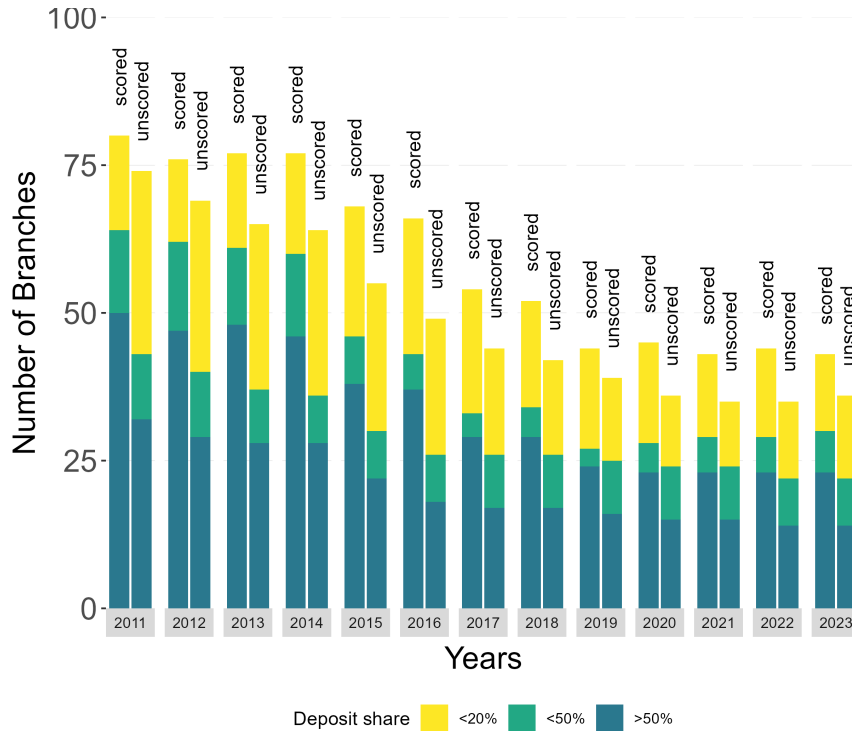


(b) Asset sizes of banks in all census tracts vs. those in scored census tracts.

**Figure 4:** Panels (a) and (b) compares Black-owned banks in all census tracts with Black-owned banks in census tracts with an HOLC score greater than 0.

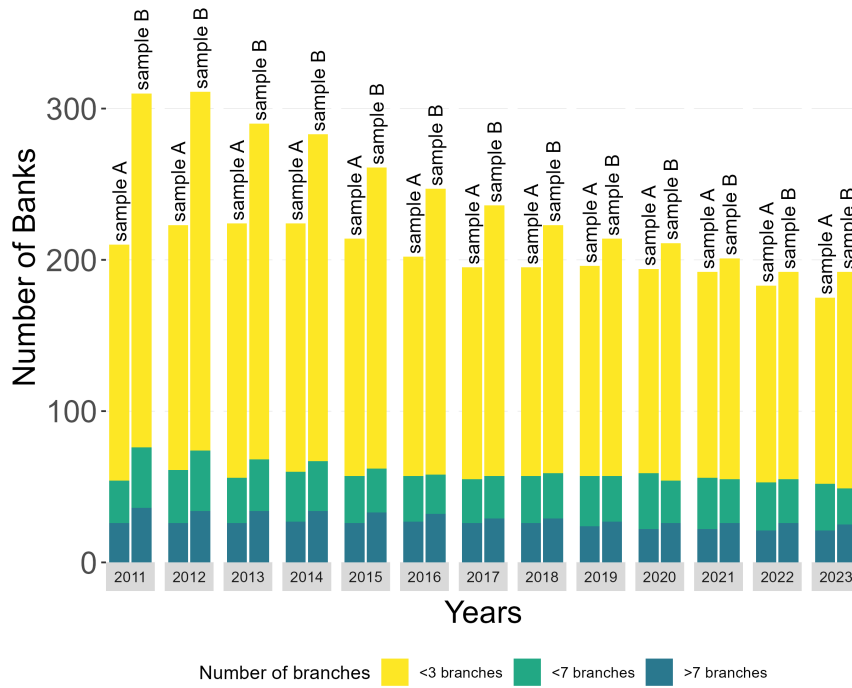


(c) Age of individual branches in all census tracts vs. those in scored census tracts.

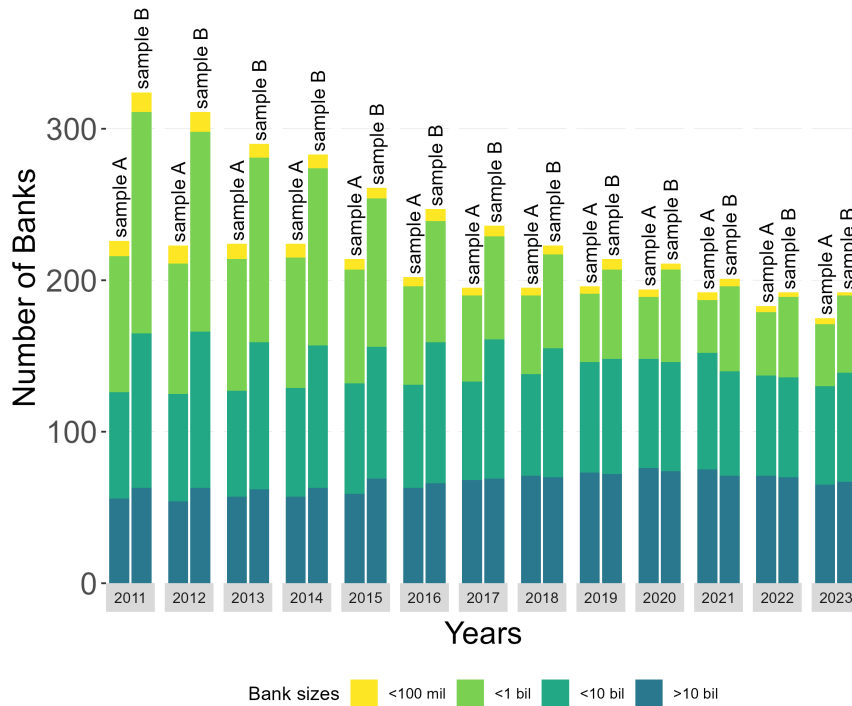


(d) Deposit share of individual branches in all census tracts vs. those in scored census tracts.

**Figure 4:** Panels (c) and (d) compares Black-owned banks in all census tracts with Black-owned banks in census tracts with an HOLC score greater than 0.



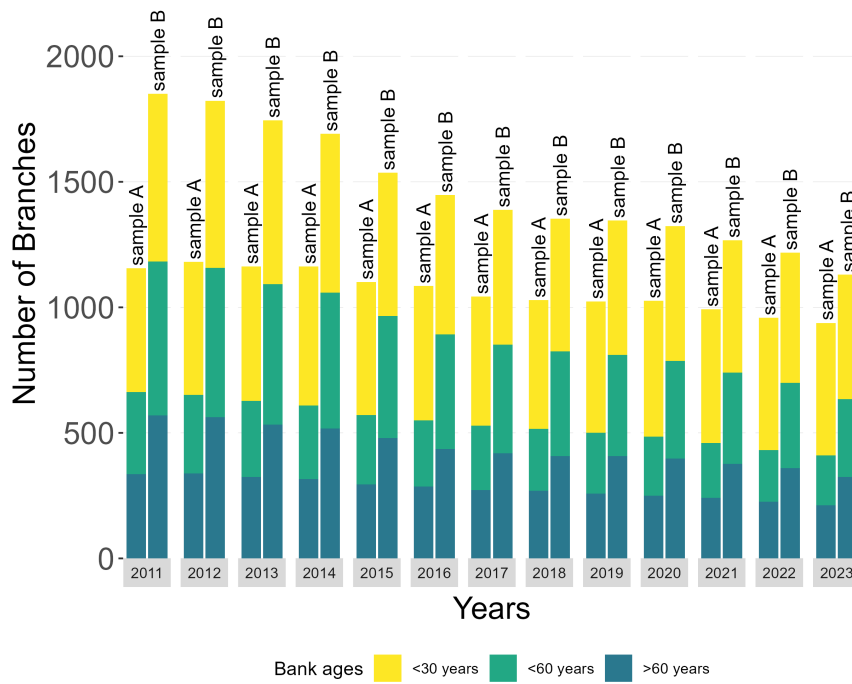
(a) Branch composition among neighboring banks of Black banks vs. non-MDI banks serving in primarily Black neighborhoods.



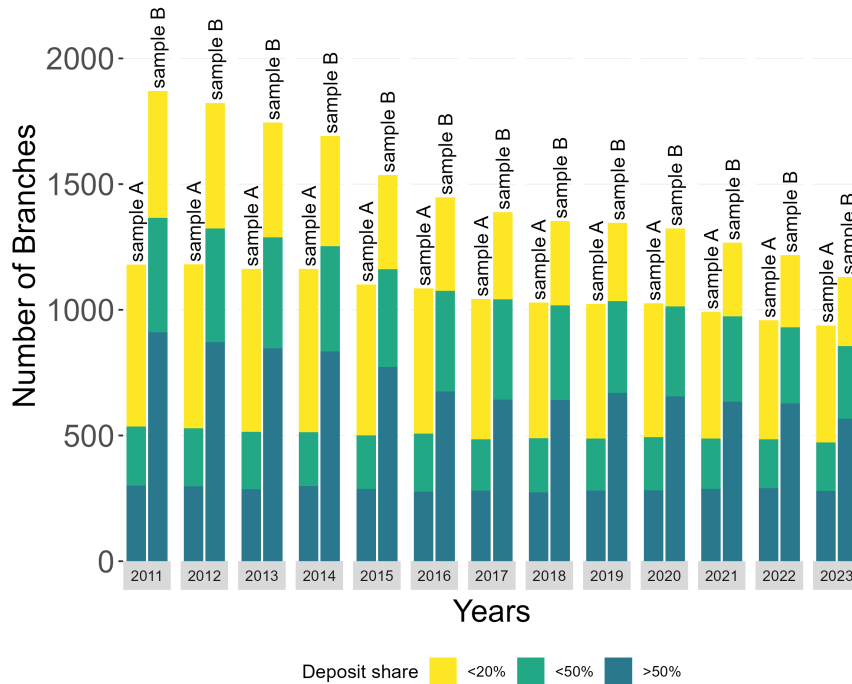
(b) Asset sizes of neighboring banks of Black banks vs. non-MDI banks serving in primarily Black neighborhoods.

**Figure 5:** Panel (a) and (b) compare neighboring banks of Black-owned banks with non-MDI banks that serve primarily Black communities, but are not neighbors of Black-owned banks.



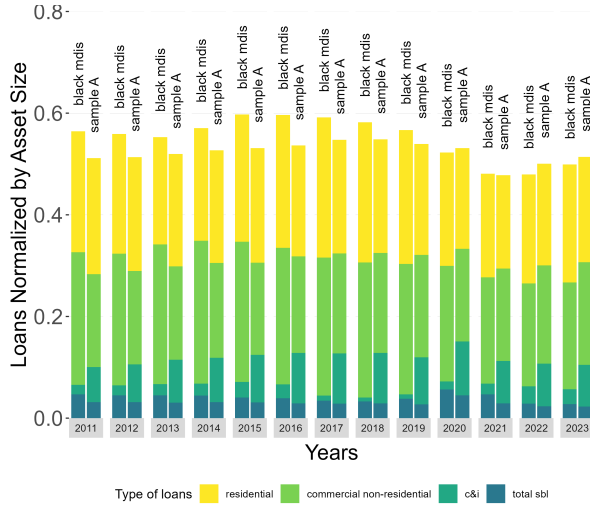


(c) Age of neighboring banks of Black banks vs. non-MDI branches serving in primarily Black neighborhoods.

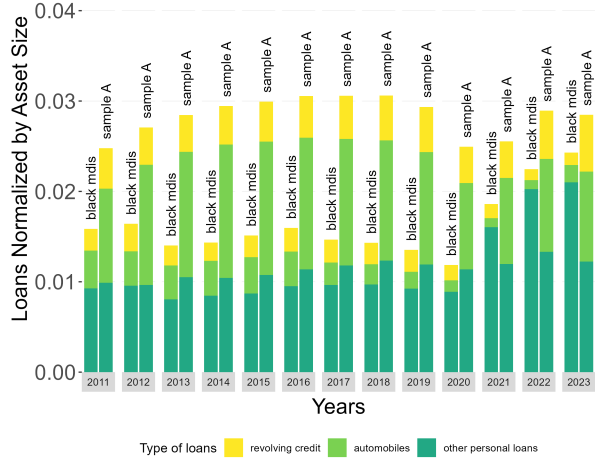


(d) Deposit share of neighboring banks of Black banks vs. non-MDI branches serving in primarily Black neighborhoods.

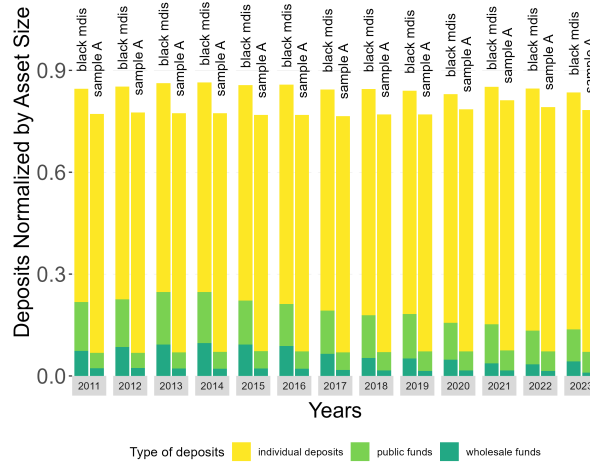
**Figure 5:** Panel (c) and (d) compare neighboring bank branches of Black-owned banks with non-MDI bank branches that serve primarily Black communities, but are not neighbors of Black-owned banks.



(a) Residential, CRE, C&I and total SBL.

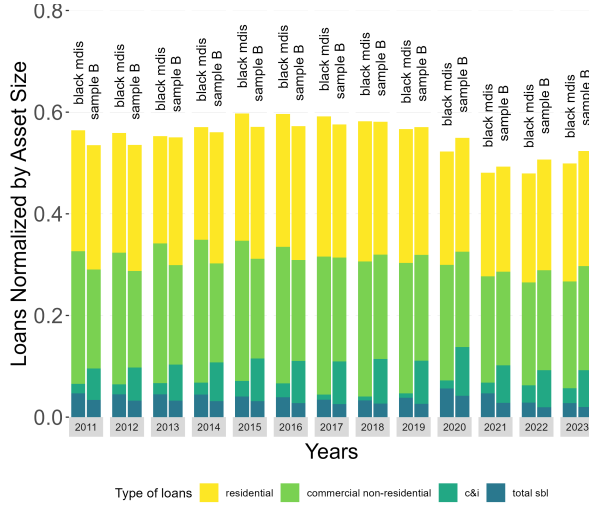


(b) Credit card and revolving credit, automobiles and other personal loans (student loans).

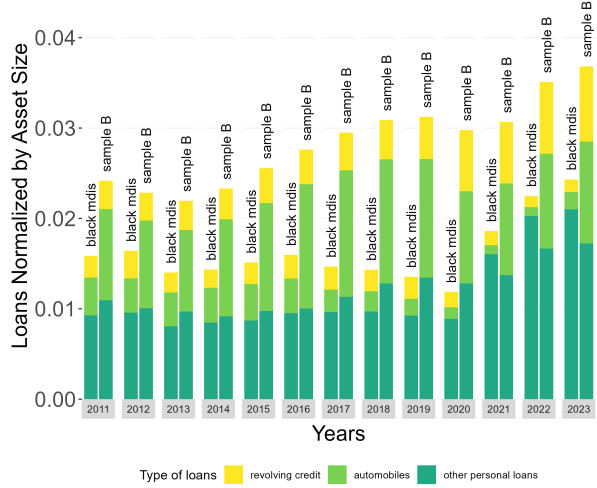


(c) Both transnational and non-transactional accounts, public funds consists of deposits from federal or state institutions.

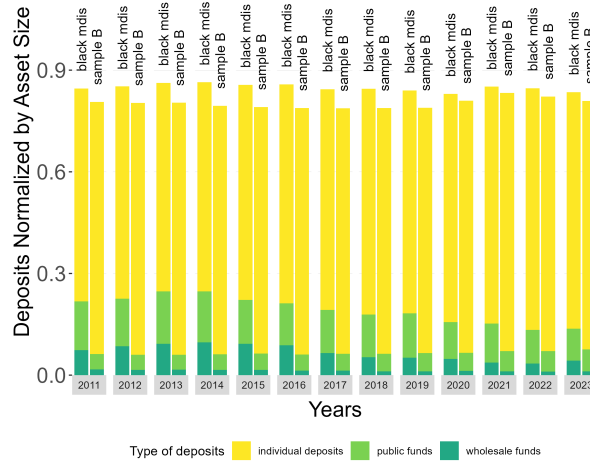
**Figure 6:** Panels (a) - (c) compare Black-owned banks with other banks in the near vicinity.



(a) Comparison of large loans- residential, CRE, C&I and total SBL.



(b) Comparison of personal loans - credit card and revolving credit, automobiles and other personal loans such as student loans.



(c) Comparison of deposit share - contains both transnational and non-transactional accounts. Public funds consists of deposits from federal or state institutions.

**Figure 7:** Panels (a) - (c) compare Black-owned banks with other banks serving primarily Black communities.

## **A Appendix: Summary Statistics of the studied tracts, in counties with Black MDIs, counties with Asian MDIs, counties with Hispanic MDIs and all counties with MDIs**

Summary statistics using 2011 as the base year for all kinds of MDIs and non-MDIs in my sample. Panel A show summary statistics for tracts in Black MDI counties, whereas Panel B through Panel D show statistics for tracts in Asian MDIs, Hispanic MDIs and non-MDIs counties respectively.

Panel A show Black MDI tracts had a higher proportion of Black population across all graded tracts and even across tracts not graded. Notably, these tracts have higher vacancies and greater proportion of renters across all grades when compared with tracts with other kinds of MDIs and non-MDIs. At the same time, median household income decreased considerably between tracts graded A and those graded D while percentage of Supplemental Nutrition Assistance Program (SNAP) beneficiary households increased.

**Table A1:** Summary statistics

|                                | <i>Panel A: Black MDI tracts</i> |                    |                    |                    |                    |
|--------------------------------|----------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                | Not Graded                       | Grade A            | Grade B            | Grade C            | Grade D            |
| Count                          | 70                               | 4                  | 11                 | 24                 | 40                 |
| Black MDIs                     | 1.07<br>(0.26)                   | 1.15<br>(0.50)     | 1.00<br>(0.001)    | 1.00<br>(0.001)    | 1.025<br>(0.001)   |
| Non MDIs                       | 3.81<br>(8.73)                   | 1.00<br>(2.00)     | 0.55<br>(0.688)    | 1.11<br>(1.41)     | 2.32<br>(4.51)     |
| Population (000s)              | 3.99<br>(1.89)                   | 3.98<br>(1.33)     | 3.79<br>(1.75)     | 3.51<br>(2.54)     | 2.28<br>(0.79)     |
| Population (000s)1 mile        | 11.770<br>(17.72)                | 23.540<br>(2.18)   | 49.120<br>(59.51)  | 29.720<br>(26.96)  | 46.523<br>(58.49)  |
| Population (000s) 5 miles      | 300.40<br>(275.40)               | 807.01<br>(387.59) | 824.84<br>(893.61) | 658.29<br>(526.56) | 738.18<br>(818.15) |
| Percent Black                  | 63.41<br>(29.17)                 | 63.25<br>(41.32)   | 75.91<br>(31.57)   | 71.45<br>(30.44)   | 69.73<br>(23.75)   |
| Percent Black 1 mile           | 41.17<br>(36.91)                 | 56.25<br>(32.54)   | 77.36<br>(23.16)   | 64.50<br>(31.54)   | 67.50<br>(18.97)   |
| Percent Black 5 miles          | 51.67<br>(23.80)                 | 28.75<br>(17.88)   | 58.91<br>(16.17)   | 49.83<br>(23.24)   | 48.78<br>(16.04)   |
| Percent Asian                  | 3.54<br>(5.26)                   | 8.25<br>(11.41)    | 3.81<br>(2.36)     | 3.63<br>(3.81)     | 3.60<br>(4.77)     |
| Percent Hispanic               | 10.14<br>(13.90)                 | 5.00<br>(1.41)     | 4.64<br>(7.07)     | 16.70<br>(23.20)   | 7.90<br>(11.01)    |
| Percent White                  | 26.18<br>(24.47)                 | 37.00<br>(26.15)   | 18.27<br>(30.47)   | 16.29<br>(19.30)   | 21.15<br>(19.75)   |
| HOLC Score                     | 0.00<br>(0.00)                   | 1.33<br>(0.36)     | 2.08<br>(0.15)     | 3.01<br>(0.14)     | 3.88<br>(0.23)     |
| Average Neighbor Score 1 mile  | 0.44<br>(0.92)                   | 1.82<br>(0.34)     | 2.35<br>(0.48)     | 2.79<br>(0.80)     | 3.48<br>(0.47)     |
| Average Neighbor Score 5 miles | 0.85<br>(1.04)                   | 2.25<br>(0.71)     | 2.37<br>(0.67)     | 2.45<br>(0.64)     | 2.53<br>(0.63)     |
| Percent Graded D               | 0.27<br>(1.10)                   | 0.00<br>(0.00)     | 1.83<br>(6.06)     | 5.26<br>(8.85)     | 74.32<br>(28.81)   |
| Average Graded D 1 mile        | 4.76<br>(13.12)                  | 5.34<br>(10.65)    | 7.74<br>(8.56)     | 20.39<br>(17.55)   | 60.99<br>(22.93)   |
| Average Grade D 5 miles        | 8.46<br>(12.85)                  | 17.23<br>(7.13)    | 19.95<br>(12.55)   | 21.83<br>(12.11)   | 28.48<br>(11.43)   |
| Median Household Income (000s) | 53.56<br>(27.30)                 | 79.98<br>(47.11)   | 50.50<br>(9.92)    | 43.02<br>(18.07)   | 37.44<br>(20.35)   |
| Percent Households on SNAP     | 18.24<br>(11.17)                 | 8.85<br>(5.94)     | 17.95<br>(11.34)   | 23.80<br>(13.48)   | 28.36<br>(12.83)   |
| Percent Vacancies              | 14.19<br>(7.45)                  | 13.37<br>(7.64)    | 12.67<br>(7.62)    | 18.34<br>(11.75)   | 21.01<br>(8.79)    |
| Percent Renters                | 49.81<br>(20.76)                 | 50.53<br>(28.27)   | 60.10<br>(29.80)   | 61.11<br>(22.46)   | 73.01<br>(15.81)   |

Population variables and median household income are in thousands.

**Table A1:** Summary statistics

|                                | <i>Panel B: Asian MDI tracts</i> |                      |                      |                      |                       |
|--------------------------------|----------------------------------|----------------------|----------------------|----------------------|-----------------------|
|                                | Not Graded                       | Grade A              | Grade B              | Grade C              | Grade D               |
| Count                          | 280                              | 8                    | 26                   | 60                   | 36                    |
| Asian MDIs                     | 1.53<br>(1.16)                   | 1.50<br>(0.76)       | 1.13<br>(0.51)       | 1.67<br>(1.16)       | 2.05<br>(1.80)        |
| Non MDIs                       | 3.24<br>(5.46)                   | 3.00<br>(0.92)       | 2.38<br>(4.00)       | 2.33<br>(3.60)       | 3.28<br>(4.47)        |
| Population (000s)              | 4.718<br>(1.957)                 | 5.472<br>(1.001)     | 4.636<br>(1.350)     | 4.515<br>(1.400)     | 4.873<br>(2.584)      |
| Population (000s) 1 mile       | 16.550<br>(29.259)               | 11.881<br>(8.314)    | 47.693<br>(26.088)   | 57.340<br>(47.973)   | 73.837<br>(54.931)    |
| Population (000s) 5 miles      | 381.670<br>(345.338)             | 712.560<br>(216.809) | 874.435<br>(441.573) | 896.224<br>(502.675) | 1297.213<br>(682.557) |
| Percent Asian                  | 28.0<br>(22.70)                  | 35.25<br>(21.49)     | 36.08<br>(22.12)     | 43.68<br>(22.49)     | 42.72<br>(26.10)      |
| Percent Asian 1 mile           | 20.59<br>(20.24)                 | 24.37<br>(23.99)     | 30.04<br>(19.56)     | 37.23<br>(19.26)     | 27.67<br>(13.85)      |
| Percent Asian 5 miles          | 18.90<br>(13.16)                 | 21.88<br>(13.09)     | 18.50<br>(9.67)      | 22.47<br>(11.50)     | 14.75<br>(7.07)       |
| Percent Black                  | 7.09<br>(9.18)                   | 8.37<br>(11.61)      | 5.81<br>(8.58)       | 6.08<br>(11.57)      | 12.22<br>(18.47)      |
| Percent Hispanic               | 23.56<br>(21.15)                 | 9.25<br>(9.30)       | 18.96<br>(16.59)     | 25.88<br>(20.17)     | 20.58<br>(19.38)      |
| Percent White                  | 50.95<br>(22.57)                 | 51.75<br>(21.93)     | 45.38<br>(23.91)     | 35.35<br>(19.22)     | 33.97<br>(19.35)      |
| HOLC Score                     | 0.00<br>(0.00)                   | 1.31<br>(0.25)       | 2.17<br>(0.19)       | 2.95<br>(0.14)       | 3.86<br>(0.26)        |
| Average Neighbor Score 1 mile  | 0.25<br>(0.68)                   | 1.60<br>(0.83)       | 2.43<br>(0.45)       | 2.71<br>(0.43)       | 2.84<br>(0.83)        |
| Average Neighbor Score 5 miles | 0.44<br>(0.83)                   | 2.46<br>(0.20)       | 2.42<br>(0.54)       | 2.38<br>(0.54)       | 2.71<br>(0.67)        |
| Percent Graded D               | 0.18<br>(1.19)                   | 0.80<br>(2.25)       | 0.50<br>(1.66)       | 1.72<br>(6.08)       | 60.59<br>(28.52)      |
| Average Graded D 1 mile        | 2.89<br>(9.31)                   | 0.63<br>(1.77)       | 4.70<br>(8.57)       | 8.02<br>(11.16)      | 41.19<br>(22.91)      |
| Average Grade D 5 miles        | 4.03<br>(9.72)                   | 10.38<br>(8.92)      | 16.00<br>(7.41)      | 15.20<br>(9.90)      | 31.50<br>(14.02)      |
| Median Household Income (000s) | 84.18<br>(39.56)                 | 135.51<br>(41.52)    | 80.56<br>(28.06)     | 62.66<br>(21.58)     | 53.71<br>(34.05)      |
| Percent Households on SNAP     | 7.25<br>(6.93)                   | 4.21<br>(2.09)       | 5.26<br>(4.09)       | 7.04<br>(7.05)       | 13.86<br>(14.41)      |
| Percent Vacancies              | 8.73<br>(6.82)                   | 4.76<br>(1.38)       | 7.35<br>(3.70)       | 7.44<br>(4.94)       | 10.31<br>(9.81)       |
| Percent Renters                | 44.22<br>(24.76)                 | 22.56<br>(15.57)     | 60.12<br>(18.84)     | 65.37<br>(21.66)     | 74.60<br>(21.79)      |

Population variables and median household income are in thousands.

**Table A1:** Summary statistics

| <i>Panel C: Hispanic MDI tracts</i> |                      |                      |                       |                       |                       |
|-------------------------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|
|                                     | Not Graded           | Grade A              | Grade B               | Grade C               | Grade D               |
| Count                               | 389                  | 6                    | 18                    | 38                    | 36                    |
| Hispanic MDIs                       | 1.11<br>(0.63)       | 1.33<br>(0.82)       | 1.17<br>(0.75)        | 1.18<br>(0.51)        | 1.19<br>(1.17)        |
| Non MDIs                            | 2.62<br>(3.43)       | 5.33<br>(6.86)       | 5.00<br>(6.86)        | 2.74<br>(4.90)        | 2.69<br>(4.68)        |
| Population (000s)                   | 5.299<br>(2.925)     | 4.705<br>(1.629)     | 5.346<br>(2.334)      | 4.457<br>(2.235)      | 4.681<br>(2.164)      |
| Population (000s) 1 mile            | 6.324<br>(10.434)    | 28.894<br>(36.415)   | 77.354<br>(81.465)    | 74.269<br>(61.393)    | 86.981<br>(61.822)    |
| Population (000s) 5 miles           | 197.214<br>(214.331) | 642.007<br>(670.789) | 1185.792<br>(950.083) | 1212.228<br>(860.709) | 1463.552<br>(940.940) |
| Percent Hispanic                    | 51.13<br>(32.77)     | 49.33<br>(26.64)     | 49.17<br>(28.23)      | 58.13<br>(29.50)      | 53.72<br>(33.65)      |
| Percent Hispanic 1 mile             | 27.04<br>(35.47)     | 53.00<br>(23.38)     | 45.22<br>(29.56)      | 51.00<br>(25.13)      | 49.61<br>(30.53)      |
| Percent Hispanic 5 miles            | 47.71<br>(32.67)     | 56.33<br>(19.73)     | 49.89<br>(19.31)      | 43.74<br>(19.97)      | 39.56<br>(19.46)      |
| Percent Asian                       | 3.86<br>(7.28)       | 3.83<br>(4.67)       | 7.50<br>(11.98)       | 9.39<br>(14.02)       | 7.47<br>(8.48)        |
| Percent Black                       | 6.61<br>(11.33)      | 6.50<br>(6.32)       | 10.61<br>(16.68)      | 9.82<br>(14.00)       | 16.69<br>(21.89)      |
| Percent White                       | 77.48<br>(15.98)     | 75.50<br>(18.67)     | 59.61<br>(27.53)      | 54.24<br>(26.17)      | 53.39<br>(27.60)      |
| HOLC Score                          | 0.00<br>(0.00)       | 1.19<br>(0.25)       | 2.12<br>(0.20)        | 2.99<br>(0.12)        | 3.85<br>(0.25)        |
| Average Neighbor Score 1 mile       | 0.10<br>(0.46)       | 1.45<br>(0.63)       | 2.01<br>(0.93)        | 2.64<br>(0.64)        | 3.13<br>(0.59)        |
| Average Neighbor Score 5 miles      | 0.20<br>(0.51)       | 1.83<br>(0.68)       | 2.04<br>(0.83)        | 2.50<br>(0.60)        | 2.55<br>(0.63)        |
| Percent Graded D                    | 0.10<br>(1.07)       | 0.00<br>(0.00)       | 0.02<br>(0.09)        | 2.98<br>(5.85)        | 65.36<br>(32.90)      |
| Average Graded D 1 mile             | 0.70<br>(4.27)       | 0.40<br>(0.73)       | 14.15<br>(17.11)      | 11.71<br>(15.44)      | 47.76<br>(25.40)      |
| Average Grade D 5 miles             | 1.43<br>(4.44)       | 12.99<br>(7.57)      | 19.27<br>(11.87)      | 21.38<br>(12.26)      | 27.56<br>(11.63)      |
| Median Household Income             | 64.05<br>(33.19)     | 67.20<br>(27.59)     | 64.45<br>(29.16)      | 52.74<br>(21.36)      | 54.69<br>(28.29)      |
| Percent Households on SNAP          | 15.35<br>(12.41)     | 13.60<br>(12.75)     | 15.64<br>(14.15)      | 17.28<br>(11.10)      | 20.22<br>(13.95)      |
| Percent Vacancies                   | 12.93<br>(9.64)      | 19.45<br>(10.81)     | 12.82<br>(9.13)       | 13.89<br>(13.18)      | 10.10<br>(5.48)       |
| Percent Renters                     | 40.06<br>(21.74)     | 59.75<br>(28.50)     | 63.60<br>(21.85)      | 68.28<br>(20.88)      | 74.18<br>(15.57)      |

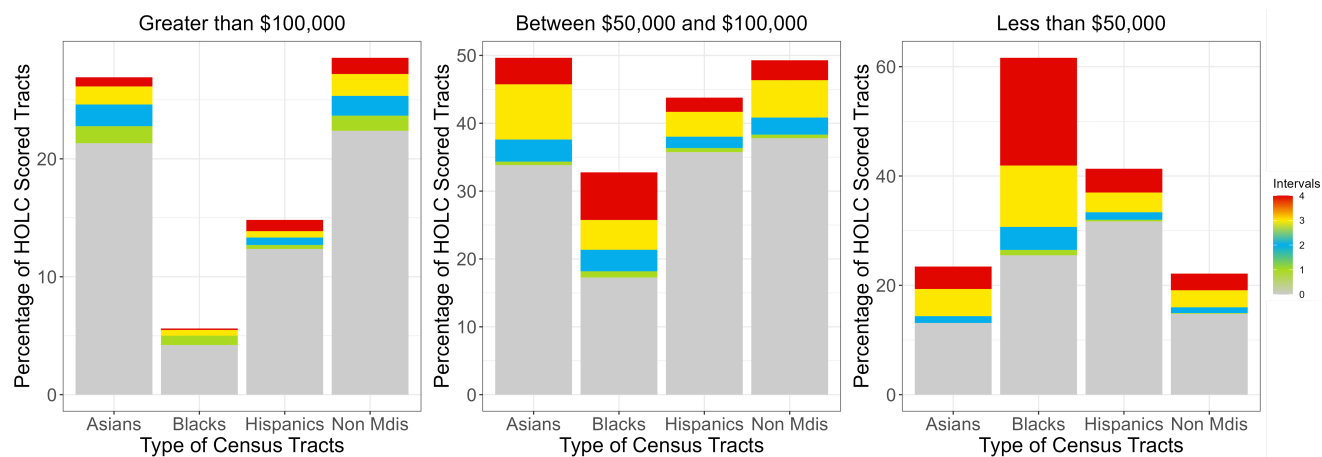
Population variables and median household income are in thousands.

**Table A1:** Summary statistics

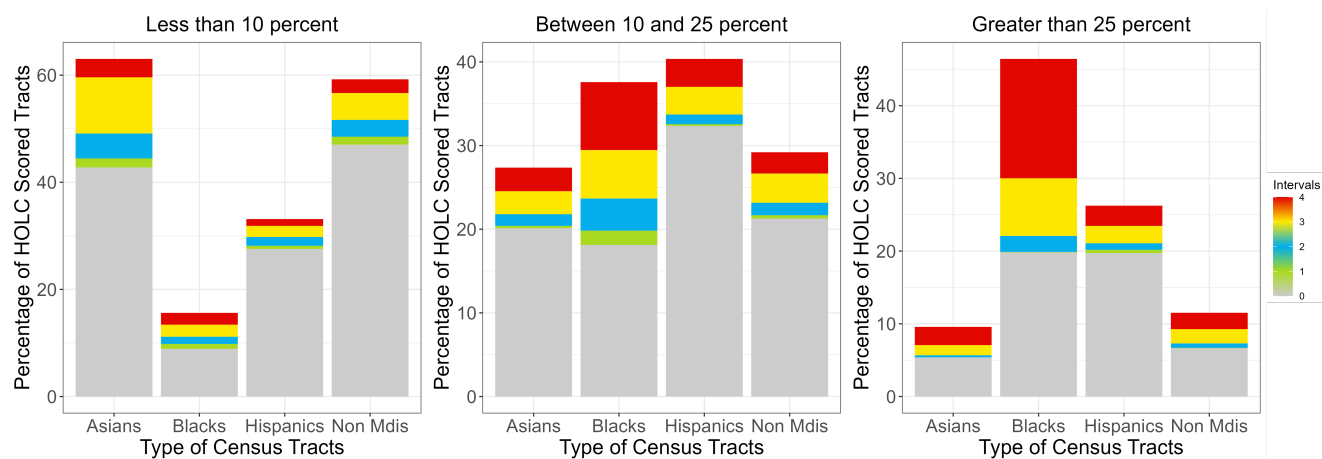
|                                | <i>Panel D: Non-MDI tracts</i> |                      |                      |                      |                      |
|--------------------------------|--------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                | Not Graded                     | Grade A              | Grade B              | Grade C              | Grade D              |
| Count                          | 24558                          | 724                  | 1680                 | 3031                 | 2216                 |
| MDIs                           | 0.00<br>(0.02)                 | 0.00<br>(0.00)       | 0.00<br>(0.00)       | 0.00<br>(0.00)       | 0.00<br>(0.00)       |
| Non MDIs                       | 2.40<br>(2.07)                 | 2.80<br>(2.82)       | 2.34<br>(2.04)       | 2.09<br>(1.78)       | 2.04<br>(2.04)       |
| Population (000s)              | 4.734<br>(2.222)               | 3.971<br>(1.477)     | 4.335<br>(1.722)     | 4.172<br>(1.750)     | 3.825<br>(2.007)     |
| Population (000s) 1 mile       | 7.643<br>(13.994)              | 26.271<br>(48.438)   | 36.609<br>(44.427)   | 40.905<br>(41.647)   | 48.888<br>(50.899)   |
| Population (000s) 5 miles      | 230.804<br>(236.150)           | 563.010<br>(640.945) | 692.048<br>(610.850) | 768.315<br>(611.357) | 899.946<br>(791.359) |
| Percent Asian                  | 7.80<br>(10.52)                | 5.38<br>(7.04)       | 7.89<br>(9.88)       | 9.25<br>(11.18)      | 8.12<br>(10.28)      |
| Percent Black                  | 12.08<br>(18.01)               | 11.57<br>(18.61)     | 17.68<br>(26.16)     | 17.61<br>(25.30)     | 23.04<br>(28.23)     |
| Percent Hispanic               | 19.16<br>(20.94)               | 9.17<br>(13.50)      | 15.43<br>(20.22)     | 24.49<br>(25.78)     | 26.40<br>(26.51)     |
| Percent White                  | 68.74<br>(22.92)               | 79.80<br>(20.20)     | 66.41<br>(26.71)     | 60.42<br>(26.53)     | 55.09<br>(26.82)     |
| Percent White 1 mile           | 40.02<br>(36.63)               | 66.12<br>(28.94)     | 59.38<br>(26.84)     | 56.64<br>(25.26)     | 62.95<br>(23.47)     |
| Percent White 5 miles          | 64.43<br>(21.01)               | 58.24<br>(15.17)     | 55.28<br>(16.13)     | 54.10<br>(15.31)     | 51.14<br>(14.58)     |
| HOLC Score                     | 0.00<br>(0.00)                 | 1.31<br>(0.25)       | 2.12<br>(0.18)       | 2.94<br>(0.15)       | 3.82<br>(0.24)       |
| Average Neighbor Score 1 mile  | 0.12<br>(0.52)                 | 1.66<br>(0.85)       | 2.05<br>(0.78)       | 2.58<br>(0.79)       | 3.00<br>(0.92)       |
| Average Neighbor Score 5 miles | 0.30<br>(0.62)                 | 1.87<br>(0.66)       | 2.12<br>(0.63)       | 2.30<br>(0.66)       | 2.50<br>(0.66)       |
| Percent Graded D               | 0.14<br>(1.31)                 | 0.45<br>(2.21)       | 0.88<br>(3.25)       | 2.87<br>(7.53)       | 62.54<br>(28.54)     |
| Average Graded D 1 mile        | 0.95<br>(6.22)                 | 5.89<br>(13.13)      | 6.29<br>(11.34)      | 11.40<br>(15.58)     | 42.76<br>(25.17)     |
| Average Grade D 5 miles        | 2.17<br>(6.37)                 | 14.30<br>(11.03)     | 15.16<br>(10.44)     | 17.34<br>(10.92)     | 26.88<br>(13.17)     |
| Median Household Income (000s) | 83.75<br>(40.33)               | 129.32<br>(59.75)    | 85.48<br>(41.46)     | 68.67<br>(30.06)     | 60.95<br>(32.05)     |
| Percent Households on SNAP     | 7.78<br>(7.78)                 | 5.40<br>(5.35)       | 8.85<br>(9.25)       | 12.08<br>(11.41)     | 18.81<br>(13.65)     |
| Percent Vacancies              | 9.64<br>(7.85)                 | 10.78<br>(8.66)      | 9.38<br>(7.08)       | 9.97<br>(7.02)       | 13.32<br>(8.62)      |
| Percent Renters                | 35.38<br>(23.59)               | 30.33<br>(19.61)     | 45.06<br>(23.73)     | 54.36<br>(22.83)     | 63.11<br>(20.19)     |

Population variables and median household income are in thousands.



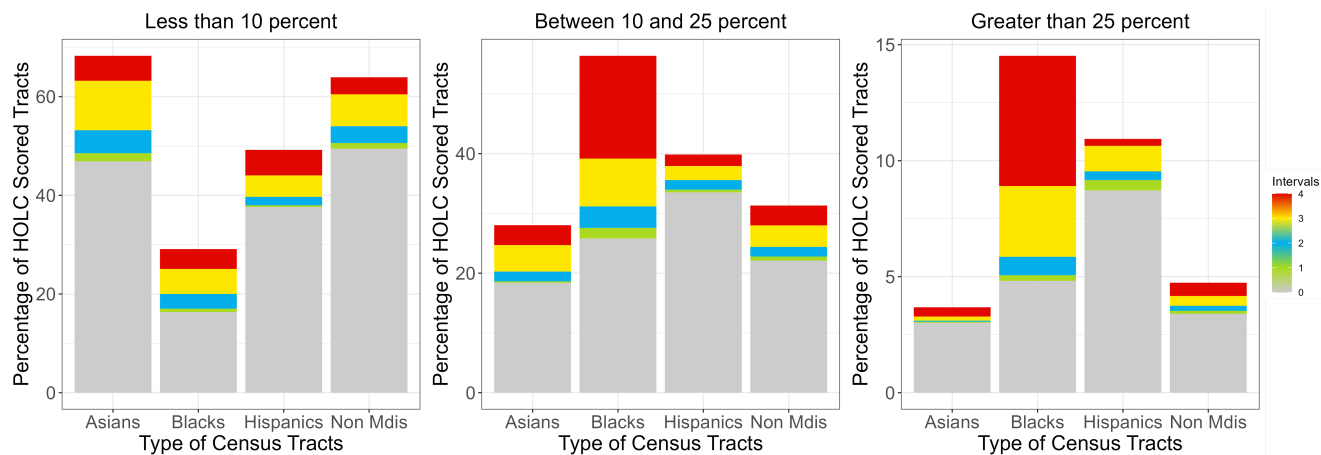


(a) Panel A - Median household income.

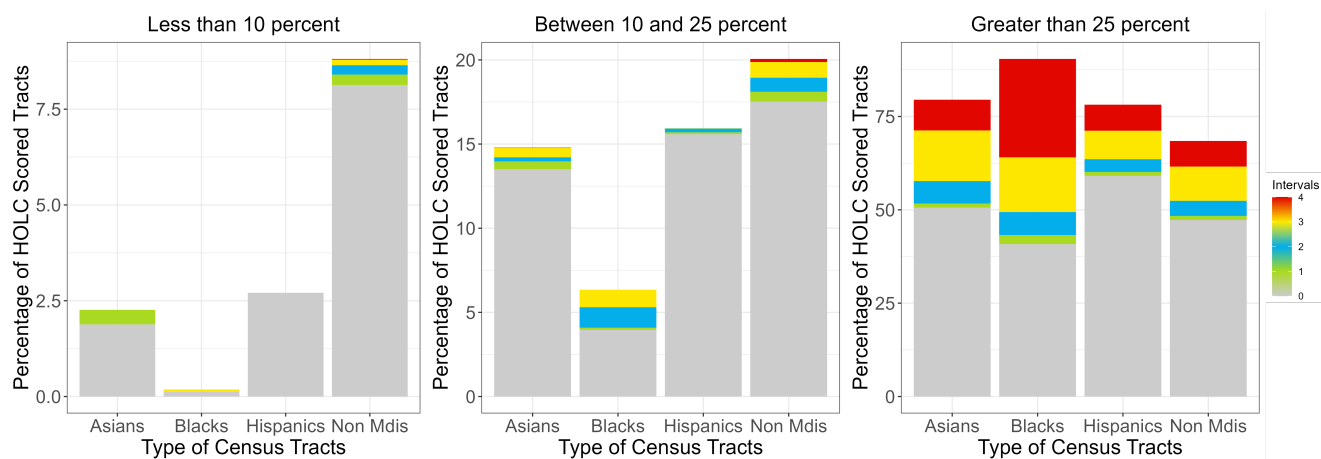


(b) Panel B - Percent of households with SNAP.

**Figure A1:** Distribution of census tracts by MDI types and percentage of HOLC scored tracts

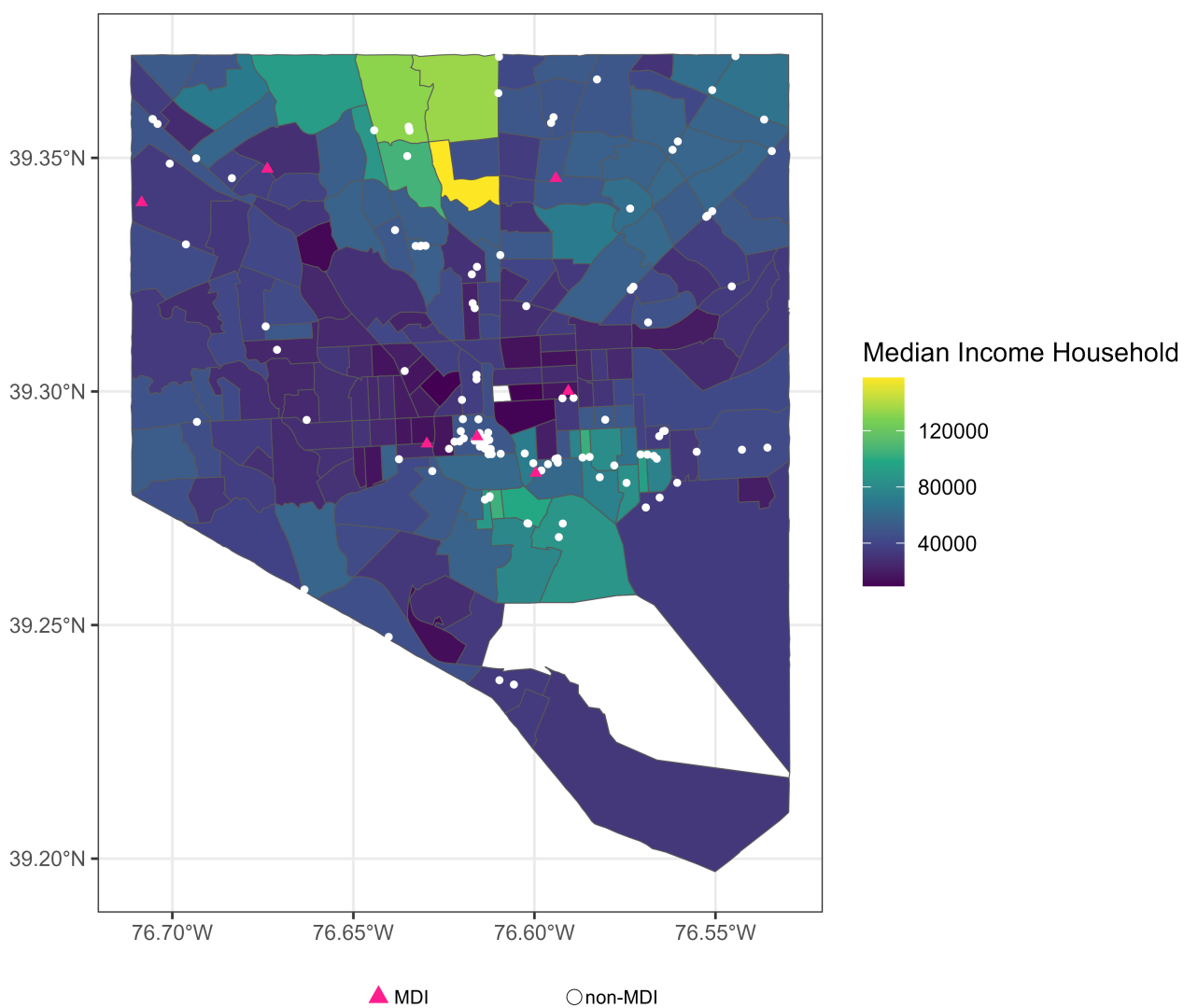


(c) Panel C - Percent of vacancies.

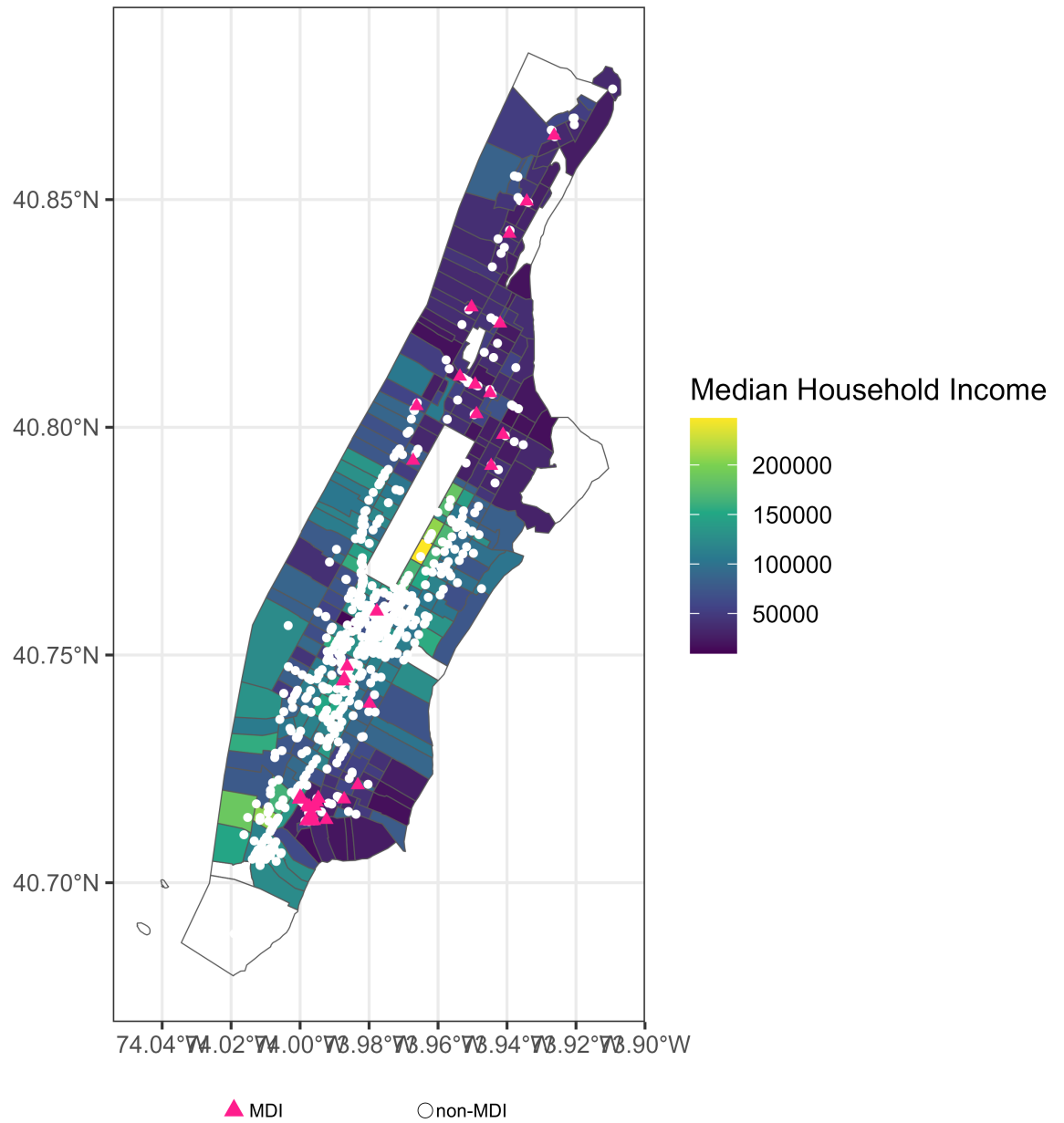


(d) Panel D - Percent of renters.

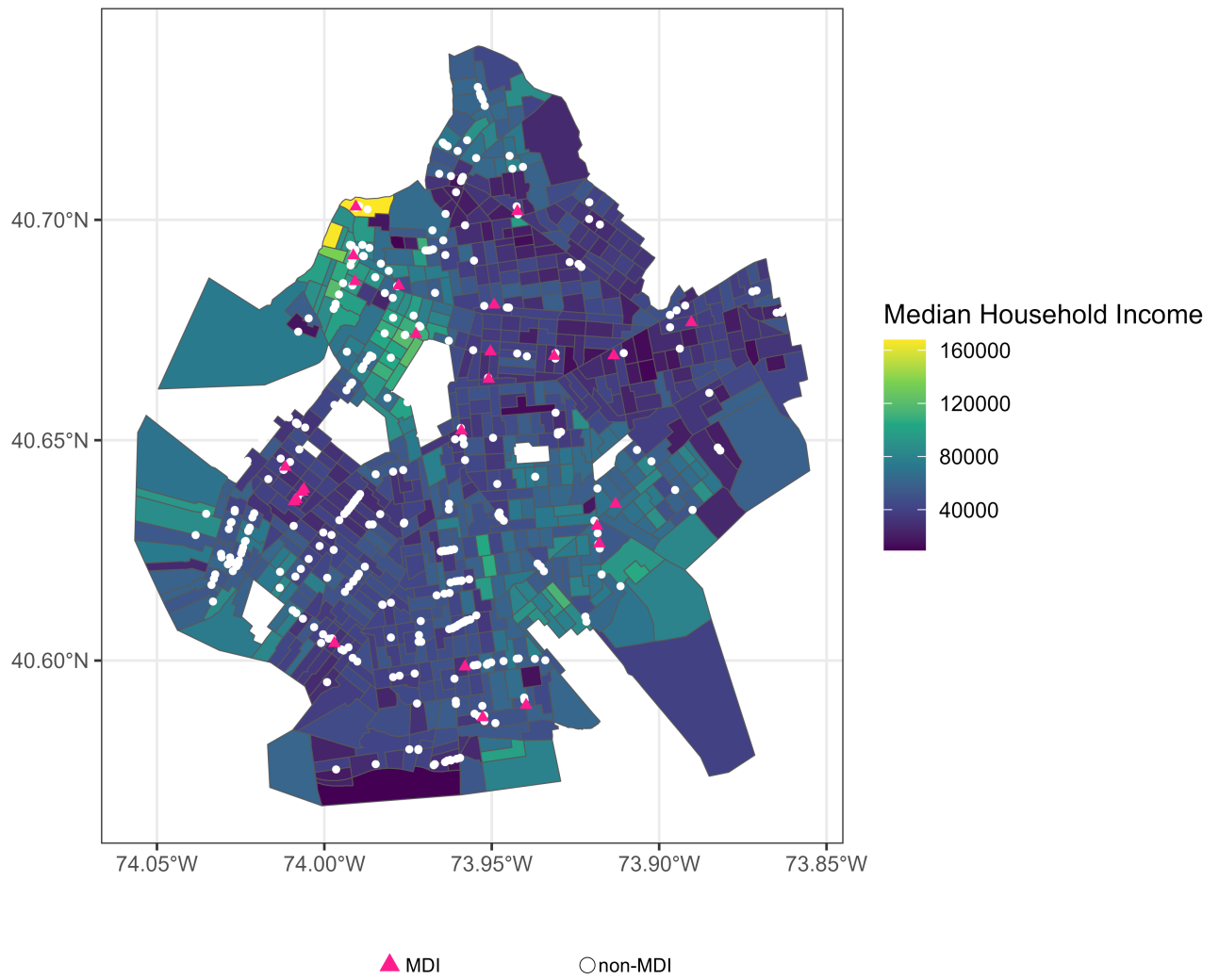
**Figure A1:** Distribution of census tracts by MDI types and percentage of HOLC scored tracts



**Figure A2:** The figure shows MDI and non-MDI branches superimposed on median household income in Baltimore city in 2011.



**Figure A3:** The figure shows poverty rates, MDI and non-MDI branches superimposed on median household income in New York county (Manhattan) in 2011.



**Figure A4:** The figure shows poverty rates, MDI and non-MDI branches superimposed on median household income in Bronx county in 2011.

## **B    Appendix: Estimation using HOLC scores or Graded D areas, sample includes all tracts with or without bank branches**

Table B1 and Table B3 show results for tracts within a mile radius. Whereas Table B2 and Table B4 show results for tracts within 5 miles radius. The results are largely similar across all specifications.

**Table B1:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −6.633***<br>(0.147) | −5.023***<br>(0.053) | −6.320***<br>(0.072) | −0.935***<br>(0.032) |
| Population (000s)                                     | 0.029**<br>(0.011)   | 0.048***<br>(0.004)  | 0.049***<br>(0.003)  | 0.065***<br>(0.002)  |
| Population (000s) 1 mile                              | 0.005***<br>(0.001)  | −0.003***<br>(0.001) | 0.001<br>(0.001)     | 0.000<br>(0.000)     |
| Percent Graded D                                      | 0.054***<br>(0.008)  | −0.020*<br>(0.009)   | −0.087***<br>(0.009) | −0.023***<br>(0.002) |
| Median Household Income                               | −0.056***<br>(0.013) | −0.049***<br>(0.005) | 0.007<br>(0.007)     | −0.010***<br>(0.001) |
| Other banks' branches                                 | 0.134***<br>(0.005)  | 0.111***<br>(0.003)  | 0.137***<br>(0.004)  |                      |
| Percent Black                                         | 0.281***<br>(0.020)  |                      |                      | −0.044***<br>(0.003) |
| Percent Black 1 mile                                  | 0.006**<br>(0.002)   |                      |                      |                      |
| Non-English Speakers                                  | −0.194***<br>(0.024) |                      |                      | 0.112***<br>(0.003)  |
| Percent Asian                                         |                      | 0.348***<br>(0.019)  |                      | 0.038***<br>(0.005)  |
| Percent Asian 1 mile                                  |                      | 0.004*<br>(0.002)    |                      |                      |
| Asian language speakers                               |                      | 0.374***<br>(0.022)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.358***<br>(0.021)  | −0.194***<br>(0.003) |
| Percent Hispanic 1 mile                               |                      |                      | −0.010***<br>(0.001) |                      |
| Percent Spanish speakers                              |                      |                      | 0.134***<br>(0.023)  |                      |
| Percent White                                         |                      |                      |                      | 0.113***<br>(0.003)  |
| Scale                                                 | 0.691                | 1.157                | 1.062                | 3.447                |
| Observations                                          | 367017               | 367017               | 367017               | 367017               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within one mile.

**Table B2:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −7.213***<br>(0.152) | −5.291***<br>(0.050) | −6.565***<br>(0.076) | −0.946***<br>(0.032) |
| Population (000s)                                     | 0.017<br>(0.012)     | 0.054***<br>(0.004)  | 0.058***<br>(0.003)  | 0.065***<br>(0.002)  |
| Population (000s) 5 miles                             | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000<br>(0.000)     | 0.000***<br>(0.000)  |
| Percent Graded D                                      | 0.067***<br>(0.008)  | −0.032***<br>(0.009) | −0.066***<br>(0.008) | −0.027***<br>(0.002) |
| Median Household Income                               | −0.060***<br>(0.013) | −0.049***<br>(0.005) | −0.017*<br>(0.007)   | −0.010***<br>(0.001) |
| Other banks' branches                                 | 0.131***<br>(0.005)  | 0.110***<br>(0.003)  | 0.133***<br>(0.004)  |                      |
| Percent Black                                         | 0.170***<br>(0.018)  |                      |                      | −0.044***<br>(0.003) |
| Percent Black 5 miles                                 | 0.031***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                  | −0.149***<br>(0.023) |                      |                      | 0.105***<br>(0.003)  |
| Percent Asian                                         |                      | 0.248***<br>(0.020)  |                      | 0.041***<br>(0.005)  |
| Percent Asian 5 miles                                 |                      | 0.019***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                |                      | 0.411***<br>(0.022)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.027<br>(0.024)     | −0.190***<br>(0.003) |
| Percent Hispanic 5 miles                              |                      |                      | 0.038***<br>(0.002)  |                      |
| Percent Spanish speakers                              |                      |                      | 0.110***<br>(0.022)  |                      |
| Percent White                                         |                      |                      |                      | 0.114***<br>(0.003)  |
| Scale                                                 | 0.615                | 1.116                | 1.182                | 3.442                |
| Observations                                          | 367017               | 367017               | 367017               | 367017               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within 5 miles.



**Table B3:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −6.580***<br>(0.145) | −5.018***<br>(0.055) | −6.220***<br>(0.072) | −0.924***<br>(0.032) |
| Population (000s)                                     | 0.027*<br>(0.011)    | 0.048***<br>(0.004)  | 0.047***<br>(0.004)  | 0.064***<br>(0.002)  |
| Population (000s) 1 mile                              | 0.005***<br>(0.001)  | −0.003***<br>(0.001) | 0.004***<br>(0.001)  | 0.000*<br>(0.000)    |
| Percent Graded D                                      | 0.079***<br>(0.023)  | −0.026<br>(0.015)    | −0.240***<br>(0.013) | −0.042***<br>(0.004) |
| Median Household Income                               | −0.062***<br>(0.013) | −0.049***<br>(0.005) | 0.004<br>(0.007)     | −0.011***<br>(0.001) |
| Other banks' branches                                 | 0.135***<br>(0.005)  | 0.111***<br>(0.003)  | 0.133***<br>(0.004)  |                      |
| Percent Black                                         | 0.282***<br>(0.020)  |                      |                      | −0.043***<br>(0.003) |
| Percent Black 1 mile                                  | 0.005**<br>(0.002)   |                      |                      |                      |
| Non-English speakers                                  | −0.201***<br>(0.025) |                      |                      | 0.118***<br>(0.003)  |
| Percent Asian                                         |                      | 0.343***<br>(0.020)  |                      | 0.035***<br>(0.005)  |
| Percent Asian 1 mile                                  |                      | 0.004**<br>(0.002)   |                      |                      |
| Percent Asian speakers                                |                      | 0.376***<br>(0.022)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.342***<br>(0.021)  | −0.199***<br>(0.003) |
| Percent Hispanic 1 mile                               |                      |                      | −0.009***<br>(0.001) |                      |
| Percent Spanish speakers                              |                      |                      | 0.147***<br>(0.023)  |                      |
| Percent White                                         |                      |                      |                      | 0.114***<br>(0.003)  |
| Scale                                                 | 0.676                | 1.150                | 1.021                | 3.435                |
| Observations                                          | 367017               | 367017               | 367017               | 367017               |

\*\*\* $p < 0.001$ ; \*\* $p < 0.01$ ; \* $p < 0.05$

**Table B4:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −7.211***<br>(0.153) | −5.242***<br>(0.052) | −6.473***<br>(0.076) | −0.931***<br>(0.033) |
| Population (000s)                                     | 0.021<br>(0.012)     | 0.053***<br>(0.004)  | 0.056***<br>(0.003)  | 0.065***<br>(0.002)  |
| Population (000s) 5 miles                             | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  |
| HOLC Score                                            | 0.112***<br>(0.022)  | −0.098***<br>(0.018) | −0.221***<br>(0.013) | −0.053***<br>(0.004) |
| Median Household Income                               | −0.064***<br>(0.013) | −0.052***<br>(0.005) | −0.023***<br>(0.007) | −0.011***<br>(0.001) |
| Other banks' branches                                 | 0.132***<br>(0.005)  | 0.107***<br>(0.003)  | 0.130***<br>(0.004)  |                      |
| Percent Black                                         | 0.169***<br>(0.018)  |                      |                      | −0.043***<br>(0.003) |
| Percent Black 5 miles                                 | 0.031***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                  | −0.156***<br>(0.023) |                      |                      | 0.111***<br>(0.003)  |
| Percent Asian                                         |                      | 0.241***<br>(0.021)  |                      | 0.038***<br>(0.005)  |
| Percent Asian 5 miles                                 |                      | 0.019***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                |                      | 0.422***<br>(0.022)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.025<br>(0.024)     | −0.195***<br>(0.003) |
| Percent Hispanic 5 miles                              |                      |                      | 0.037***<br>(0.002)  |                      |
| Percent Spanish speakers                              |                      |                      | 0.116***<br>(0.022)  |                      |
| Percent White                                         |                      |                      |                      | 0.115***<br>(0.003)  |
| Scale                                                 | 0.604                | 1.075                | 1.131                | 3.430                |
| Observations                                          | 367017               | 367017               | 367017               | 367017               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within 5 miles.

## **C    Appendix: Estimation using average HOLC scores or average Graded D**

I calculate the average HOLC scores and average percentage of tracts Graded D of neighboring tracts within a mile and 5 miles respectively and use them as the primary explanatory variable. Table C1 and Table C3 show results for tracts within a mile radius. Whereas Table C2 and Table C4 show results for tracts within 5 miles radius. The results are largely similar across all specifications.

**Table C1:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −6.646***<br>(0.113) | −4.660***<br>(0.040) | −5.404***<br>(0.043) | −0.061*<br>(0.030)   |
| Population (000s)                                     | −0.035*<br>(0.014)   | 0.019***<br>(0.005)  | 0.005<br>(0.005)     | 0.032***<br>(0.001)  |
| Population (000s) 1 mile                              | 0.006***<br>(0.001)  | −0.001*<br>(0.000)   | 0.006***<br>(0.000)  | 0.000**<br>(0.000)   |
| Average Percent Graded D                              | 0.169***<br>(0.010)  | −0.005<br>(0.010)    | −0.082***<br>(0.011) | 0.032***<br>(0.002)  |
| Median Household Income                               | −0.035**<br>(0.012)  | −0.043***<br>(0.004) | 0.006<br>(0.007)     | 0.006***<br>(0.001)  |
| Other banks' branches                                 | 0.107***<br>(0.004)  | 0.082***<br>(0.003)  | 0.107***<br>(0.003)  |                      |
| Percent Black                                         | 0.384***<br>(0.018)  |                      |                      | 0.005<br>(0.003)     |
| Percent Black 1 mile                                  | 0.007***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                  | −0.135***<br>(0.024) |                      |                      | 0.094***<br>(0.003)  |
| Percent Asian                                         |                      | 0.328***<br>(0.022)  |                      | 0.037***<br>(0.005)  |
| Percent Asian 1 mile                                  |                      | 0.008***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                |                      | 0.293***<br>(0.023)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.456***<br>(0.017)  | −0.117***<br>(0.003) |
| Percent Hispanic 1 mile                               |                      |                      | −0.009***<br>(0.001) |                      |
| Percent Spanish speakers                              |                      |                      | 0.117***<br>(0.021)  |                      |
| Percent White                                         |                      |                      |                      | 0.070***<br>(0.003)  |
| Scale                                                 | 0.645                | 1.106                | 1.083                | 2.693                |
| Observations                                          | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within one mile.

**Table C2:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on percent graded D within census tracts</i> |                      |                      |                      |                      |
|-------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                       | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                           | −6.833***<br>(0.112) | −4.907***<br>(0.041) | −5.833***<br>(0.052) | −0.062*<br>(0.029)   |
| Population (000s)                                     | −0.089***<br>(0.016) | 0.018***<br>(0.005)  | 0.010*<br>(0.005)    | 0.033***<br>(0.001)  |
| Population (000s) 5 miles                             | 0.001***<br>(0.000)  | 0.000***<br>(0.000)  | 0.001***<br>(0.000)  | −0.000<br>(0.000)    |
| Average Percent Graded D                              | 0.130***<br>(0.025)  | −0.075***<br>(0.023) | −0.319***<br>(0.020) | 0.078***<br>(0.005)  |
| Median Household Income                               | −0.071***<br>(0.013) | −0.054***<br>(0.004) | −0.034***<br>(0.007) | 0.006***<br>(0.001)  |
| Other banks' branches                                 | 0.101***<br>(0.005)  | 0.085***<br>(0.003)  | 0.111***<br>(0.003)  |                      |
| Percent Black                                         | 0.308***<br>(0.018)  |                      |                      | 0.001<br>(0.003)     |
| Percent Black 5 miles                                 | 0.022***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                  | −0.119***<br>(0.023) |                      |                      | 0.092***<br>(0.003)  |
| Percent Asian                                         |                      | 0.202***<br>(0.024)  |                      | 0.035***<br>(0.005)  |
| Percent Asian 5 miles                                 |                      | 0.023***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                |                      | 0.365***<br>(0.024)  |                      |                      |
| Percent Hispanic                                      |                      |                      | 0.218***<br>(0.023)  | −0.114***<br>(0.003) |
| Percent Hispanics 5 miles                             |                      |                      | 0.028***<br>(0.002)  |                      |
| Percent Spanish speakers                              |                      |                      | 0.075***<br>(0.019)  |                      |
| Percent White                                         |                      |                      |                      | 0.068***<br>(0.003)  |
| Scale                                                 | 0.545                | 1.177                | 1.126                | 2.659                |
| Observations                                          | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within five miles.

**Table C3:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on HOLC Score using Meier and Mitchell (2023)</i> |                      |                      |                      |                      |
|------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                            | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                                | −6.476***<br>(0.114) | −4.672***<br>(0.041) | −5.337***<br>(0.044) | −0.055*<br>(0.028)   |
| Population (000s)                                          | −0.061***<br>(0.015) | 0.019***<br>(0.005)  | 0.001<br>(0.005)     | 0.032***<br>(0.001)  |
| Population (000s) 1 mile                                   | 0.007***<br>(0.001)  | −0.001**<br>(0.001)  | 0.009***<br>(0.000)  | 0.001***<br>(0.000)  |
| Average HOLC Score                                         | 0.184***<br>(0.024)  | 0.018<br>(0.014)     | −0.287***<br>(0.017) | 0.024***<br>(0.004)  |
| Median Household Income                                    | −0.054***<br>(0.012) | −0.042***<br>(0.004) | 0.003<br>(0.006)     | 0.006***<br>(0.001)  |
| Other banks' branches                                      | 0.107***<br>(0.004)  | 0.082***<br>(0.003)  | 0.108***<br>(0.003)  |                      |
| Percent Black                                              | 0.380***<br>(0.019)  |                      |                      | 0.005<br>(0.003)     |
| Percent Black 1 mile                                       | 0.005**<br>(0.002)   |                      |                      |                      |
| Non-English speakers                                       | −0.162***<br>(0.024) |                      |                      | 0.089***<br>(0.003)  |
| Percent Asian                                              |                      | 0.329***<br>(0.021)  |                      | 0.038***<br>(0.004)  |
| Percent Asian 1 mile                                       |                      | 0.008***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                     |                      | 0.289***<br>(0.023)  |                      |                      |
| Percent Hispanic                                           |                      |                      | 0.445***<br>(0.017)  | −0.114***<br>(0.003) |
| Percent Hispanic 1 mile                                    |                      |                      | −0.008***<br>(0.001) |                      |
| Percent Spanish speakers                                   |                      |                      | 0.126***<br>(0.020)  |                      |
| Percent White                                              |                      |                      |                      | 0.069***<br>(0.003)  |
| Scale                                                      | 0.622                | 1.106                | 1.065                | 2.707                |
| Observations                                               | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within one mile.

**Table C4:** Poisson GEE estimates on the count of MDI and non-MDI branches

| <i>Based on HOLC Score using Meier and Mitchell (2023)</i> |                      |                      |                      |                      |
|------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
|                                                            | Black MDIs           | Asian MDIs           | Hispanic MDIs        | Other Bank Branches  |
| (Intercept)                                                | −6.854***<br>(0.113) | −4.886***<br>(0.043) | −5.745***<br>(0.052) | −0.081**<br>(0.028)  |
| Population (000s)                                          | −0.089***<br>(0.016) | 0.017***<br>(0.005)  | 0.006<br>(0.005)     | 0.033***<br>(0.001)  |
| Population (000s) 5 miles                                  | 0.001***<br>(0.000)  | 0.000***<br>(0.000)  | 0.001***<br>(0.000)  | 0.000*<br>(0.000)    |
| Average HOLC Score                                         | 0.139***<br>(0.032)  | −0.084***<br>(0.023) | −0.433***<br>(0.023) | 0.063***<br>(0.005)  |
| Median Household Income                                    | −0.076***<br>(0.013) | −0.055***<br>(0.004) | −0.037***<br>(0.007) | 0.006***<br>(0.001)  |
| Other banks' branches                                      | 0.104***<br>(0.004)  | 0.084***<br>(0.003)  | 0.110***<br>(0.003)  |                      |
| Percent Black                                              | 0.302***<br>(0.018)  |                      |                      | 0.002<br>(0.003)     |
| Percent Black 5 miles                                      | 0.023***<br>(0.002)  |                      |                      |                      |
| Non-English speakers                                       | −0.131***<br>(0.023) |                      |                      | 0.081***<br>(0.003)  |
| Percent Asian                                              |                      | 0.187***<br>(0.024)  |                      | 0.042***<br>(0.005)  |
| Percent Asian 5 miles                                      |                      | 0.024***<br>(0.002)  |                      |                      |
| Percent Asian speakers                                     |                      | 0.380***<br>(0.025)  |                      |                      |
| Percent Hispanic                                           |                      |                      | 0.214***<br>(0.023)  | −0.107***<br>(0.003) |
| Percent Hispanics 5 miles                                  |                      |                      | 0.028***<br>(0.002)  |                      |
| Percent Spanish speakers                                   |                      |                      | 0.080***<br>(0.019)  |                      |
| Percent White                                              |                      |                      |                      | 0.069***<br>(0.003)  |
| Scale                                                      | 0.530                | 1.164                | 1.096                | 2.685                |
| Observations                                               | 166174               | 166174               | 166174               | 166174               |

Notes: \*\*\* $p < 0.01$ ; \*\* $p < 0.05$ ; \* $p < 0.1$ 

Neighboring tracts are within five miles.